Master's thesis

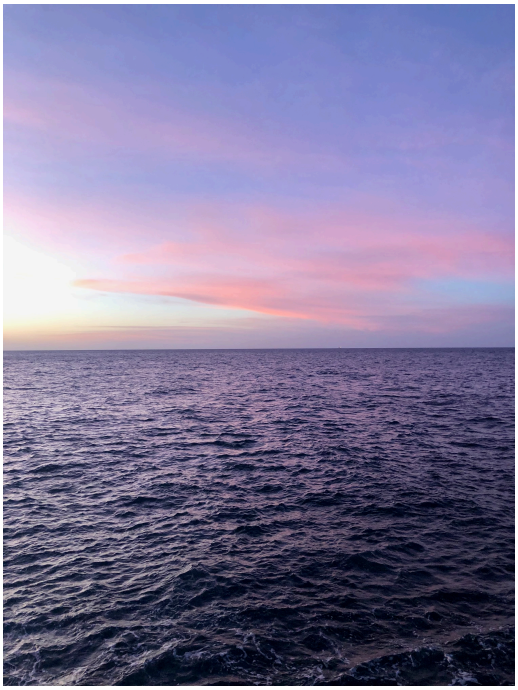
NTNU
Norwegian University of Science and Technology
Faculty of Natural Sciences
Department of Physics

Markus Woxholt

Ocean Wave-Current Interactions Including Viscosity and Shear

A first-principles analytical and numerical investigation in two dimensions

Master's thesis in Applied Physics and Mathematics
Supervisor: Lars Erik Holmedal
Co-supervisor: Dag Myrhaug, Jon Andreas Støvneng
June 2026



Det karibiske hav, januar 2023. Markus Woxholt

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ABSTRACT

Wave-current interactions (WCI) govern processes of practical importance in marine engineering and oceanography, yet many classical models neglect viscosity. This thesis develops a two-dimensional model of linear deep-water surface waves propagating on a sheared background current, derived directly from the viscous Navier-Stokes equation. The wave–current system is linearized, reduced to its WCI-coupled terms, and treated by a WKB expansion.

At leading order the system reduces to a single fourth-order ordinary differential equation for the vertical velocity, recovering the Orr–Sommerfeld equation. The leading-order equations are validated in the inviscid limit and are shown to admit a depth-dependent eddy-viscosity closure. Two background-current profiles are solved in closed form. For a depth-invariant current, the dispersion relation independently derives a result similar in form to Dias et al. (2008), but expressed in the intrinsic frequency, confirming that the current enters as a Doppler shift. A higher-order correction to the classical Lamb dissipation rate is identified, which in the uniform eddy-viscosity regime, shows to reduce the maximum deviation from the full dispersion relation from approximately 3% to 0.002%.

For a linear shear current, a new dispersion relation is obtained, closed in terms of Airy-function eigenmodes, generalizing the viscous current-free result of Dias et al. (2008) and the inviscid linear-shear result of Ellingsen and Brevik (2014) to one single model. Evaluated for realistic wind-induced current parameters, this relation produces only small and well-behaved corrections to established theory, deviating from inviscid shear theory by below $\sim 10^{-2}\%$ in real frequency and correcting the depth-invariant dissipation rate by at most $\sim 0.05\%$. These corrections are physically consistent but reveal no new significant effect of the viscous-shear coupling. Finally, the second-leading-order WKB system is derived as a basis for future work on amplitude modulation.

The model establishes a first-principles framework for incorporating viscosity and shear into WCI. The viscous corrections are negligible at molecular viscosity but become appreciable for eddy viscosities representative of a turbulent surface layer. Depth-dependent eddy viscosity and coupling to larger-scale ocean and wind models are identified as immediate extensions of the research of this thesis.

SAMMENDRAG

Bølge-strøm interaksjoner (WCI) påvirker prosesser av praktisk betydning innen marin ingeniørvitenskap og oseanografi. Likevel er det vanlig at klassiske modeller neglisjerer viskositet. I denne oppgaven blir en todimensjonell modell for lineære dypvannsbølger som propagerer på en skjærstrøm utviklet direkte fra den viskøse Navier-Stokes-likningen. Bølge-strømsystemet blir linearisert, redusert slik at kun bølge-strømkoblede ledd gjenstår og utviklet i en WKB-ekspansjon.

I ledende orden reduserer systemet til en fjerdeordens ordinær differensiallikning for den vertikale hastigheten. Denne er Orr–Sommerfeldlikningen. Ledende ordens likninger blir validert for ikke-viskøse grenseverdier og det blir vist at de er kompatible med en dybdeavhengig eddy-viskositet. To bakgrunnsstrømsprofiler blir løst i lukket form. For en dybdeinvariant strøm gir dispersjonsrelasjonen et uavhengig resultat likt Dias et al. (2008) i formen, men gitt som en funksjon av egenfrekvensen. Dette bekrefter at strømmen entrer systemet som et dopplerskift. En $\mathcal{O}(\nu^{3/2})$-korreksjon til den klassiske Lamb-dissipasjonsraten blir funnet. Korreksjonen reduserer det maksimale avviket fra den fulle dispersjonsrelasjonen fra nesten 3% til 0.002%.

For en lineær skjærstrøm blir en ny dispersjonsrelasjon utledet. Den har lukket form uttrykt i Airyfunksjonsegenmoder og generaliserer det viskøse nullstrømsresultatet til Dias et al. (2008) og Ellingsen og Breviks (2014) ikke-viskøse resultat for lineært skjær til én modell. Evaluert for realistiske parametere for vinddrevet strøm gir denne relasjonen små korreksjoner til standard teori. Den avviker fra ikke-viskøs skjærstrøm med under $\sim 10^{-2}\%$ for frekvens og korrigerer den dybdeinvariante dissipasjonsraten med opp til $\sim 0.05\%$. Disse korreksjonene er fysisk konsistente, men avdekker ingen ny signifikant effekt av koblet viskositet og skjær. Til slutt blir WKB-system utledet til andre orden for å danne et grunnlag for fremtidig arbeid med å finne moderering av amplituden.

Modellen etablerer et grunnprinsippsrammeverk for å inkorporere viskositet og skjær i WCI. De viskøse korreksjonene er neglisjerbare for molekylær viskositet, men er signifikante for eddyviskositeter representative for turbulente overflategrensesjikt. Å inkludere dybdeavhengig eddyviskositet og kobling til storskala hav- og vindmodeller blir identifisert som umiddelbare videreføringer av forskningen i denne oppgaven.

ACKNOWLEDGEMENTS

This master's thesis is the final result of the work done in TFY4510 and TFY4900 at the Norwegian University of Science and Technology in Trondheim. The project was initiated during the spring semester of 2025 and finalized in June 2026. I would like to thank my supervisor, Professor Lars Erik Holmedal at the NTNU's Department of Marine Technology, whose insights and pedagogical clarity have been instrumental in shaping the direction of the project. I extend my gratitude to Professor Emeritus Dag Myrhaug for his guidance and continuous support throughout the course of the project. I will also thank my department supervisor, Associate Professor Jon Andreas Støvneng for organizing the practical issues of concern to this master's thesis.

Finally, I will send a big thank you to my family and all my friends in Trondheim and at NTNU who have made my study years such a fun and memorable time.

Declaration on the Use of Artificial Intelligence

During the development of this thesis, generative artificial intelligence (AI) have been used as a supporting tool. The AI's used are exclusively Claude AI by Anthropic and ChatGPT by OpenAI, which we henceforth refer to as *the AI tools*. The AI tools have been used to the following ends:

- **Numerical solver development and plotting:** The AI tools were used to help write numerical solvers in Python. The code foundations were written by me, and all AI-suggested edits have been proofread before inclusion. All numerical results have been thoroughly checked and validated by me.
- **Thesis structure and language:** The written words of this thesis are my own and I have in cooperation with my supervisors planned and executed the project. However, the AI tools have been used to suggest improvements in the precision of language and the structure within sections of the thesis.
- **Academic discussion:** All reasoning, discussion and conclusion in the thesis are my own. The AI tools were critically used as a discussion partner and did occasionally help identify relevant concepts or literature.
- **Drawing of figures:** The AI tools have been used to draw the illustrative figures in this thesis based on my written descriptions. This does not extend

to the result figures, which are numerical solutions of the physics presented explicitly in the thesis.

Declaration of Similarities with Previous Work

I wish to highlight the sections of this thesis which have similarities or are reused from the preliminary project assignment-course TFY4510:

- The structure and language of the first subsections of §2.2 follow closely that of the project assignment. However, most essential equations are changed and starting from §2.2.2.2 there is no content overlap, but the structure is similar.
- §2.3 has large sections of overlap but the equations are updated and language refined.
- §2.5 has sections of overlap, but a figure is changed and language is refined.
- The introductory paragraph of §2.6 is copied from the project assignment.
- The structure and language of §3 overlaps, but new paragraphs and a new figure have been added. All equations and physical descriptions, except (3.19) and (3.3), are changed.
- A different version of (3.14) was presented in the project assignment.
- §3.2 and §4.2.2 are modified parts of what was one subsection in the project assignment.

Declaration of Compliance

I declare that I have carried out this work independently and in compliance with the exam regulations of the Norwegian University of Science and Technology.

Markus Woxholt
Trondheim
June 19, 2026

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ABBREVIATIONS

List of all abbreviations in alphabetic order:

- **WCI** Wave-current interactions
- **WKB** Wentzel–Kramers–Brillouin.
- **ODE** Ordinary differential equations
- **PDE** Partial differential equations

CHAPTER

ONE

INTRODUCTION

1.1 Motivation

The interplay between surface waves and ocean currents, commonly referred to as wave–current interaction (WCI), is important in coastal engineering and oceanography as it influences sediment transport, hydrodynamic forces on offshore structures, pollutant dispersion, and wave transformation. Many classical models typically neglect viscosity in the derivation of wave kinematics and dynamics, even though in a real fluid, viscosity introduces internal shear stresses, enables boundary layer development, and affects energy dissipation. Incorporating viscosity into wave models could thus improve model accuracy.

A complicating factor in WCI models is that ocean currents, whether geostrophic, tidal, or wind induced, are turbulent in nature. The complete understanding of turbulence is one of the great challenges of modern mathematical physics and fluid mechanics, and incorporating viscosity in WCI models is only a first step towards accurately describing the inherent mechanisms of these complex interactions. Sir Horace Lamb, one of the most significant contributors to fluid mechanics reputedly said on his death bed: "I am an old man now, and when I die and go to heaven there are two matters on which I hope for enlightenment. One is quantum electrodynamics, and the other is the turbulent motion of fluids. And about the former I am rather optimistic.". In other words, including turbulence precisely in WCI models is no small task.

To this day, even the simplest turbulence models lead to a great depth of complexity when included in WCI models, and new insights can be attained by investigating relatively plain configurations. A widely used approach is to include turbulence via the viscosity, often using an eddy viscosity. The work of this thesis is motivated by the need for better insight into the complex interactions between shear, viscosity, and turbulence in WCI modeling, and the potentially fruitful rewards even small insights into this relatively unknown area could bring.

1.2 Literature Review

The subject of WCI has been thoroughly studied over the past century and a half, following foundational contributions by Stokes, Lamb, and others. For a long period the theory remained fragmented, presented mainly as separate findings within the broader field of fluid dynamics, rather than as a coherent discipline. Peregrine's work through the 1970s was among the first to consolidate the separate strands of WCI theory into a unified framework. Writing for the U.S. Army Coastal Engineering Research Center, Peregrine et al. (1983) note that although WCI had received relatively more attention in Northern Europe, it remained underestimated in the American engineering community. This, they write, despite the fact that the observed behavior of waves in the presence of currents differs substantially from predictions that treat waves and currents as independent entities. The report summarizes the theoretical foundations for current's effects on waves, covering the dispersion relation, wave breaking, refraction, diffraction, and dissipation.

In the decades since, the field has grown large and increasingly applied. Zhang et al. (2022) provide a more recent comprehensive review of WCI research across analytical, numerical, and experimental perspectives. Much of this work proceeds within established frameworks and empirical closures rather than by re-examining foundations. Zhang et al. conclude that changes in wave characteristics such as wavelength and wave height in the upper layers of the water column are well predicted by existing models. However, they point out that turbulent viscosity models are less convincingly validated in the upper layers. This shows that the contemporary literature has advanced the applied modeling of WCI, but there is room for re-examining its viscous foundations. We intend for this thesis to contribute to this end and will subsequently present literature that helps to locate this thesis in the basic research tradition of WCI.

Simmen (1984) studies steady deep-water waves on a linear shear current via a variational principle and numerical continuation, extending the analysis to large-amplitude and extreme waves. Ellingsen and Brevik (2014) gives a thorough treatment of the inviscid linear problem on a linear shear current, noting that despite its foundational status it is rarely presented in general form in the WCI literature. Both remain inviscid; Dias et al. (2008), conversely, isolate viscous dissipation in the absence of any current, re-deriving the classical $\mathcal{O}(\nu)$ viscous dissipation result (Lamb, 1932) for deep-water waves and show that it appears naturally in the nonlinear Schrödinger equation. They close with a comment on the tendency to cite previous papers without providing a convincing physical explanation. The aforementioned papers are examples of basic research, closing gaps and extending established theory by deriving results from first principles, with the assumptions made explicit, an example we aim to follow in this thesis.

1.3 Scope and Limitations

This work is intended to lay the foundation for future development of viscous WCI models and to take a first step towards understanding how strong shear currents and turbulence influence wave propagation under realistic oceanographic settings. To narrow down this broad objective we propose the following research questions: (i) *What form does the viscous dispersion relation take when derived directly from a coupled wave–current system*, and (ii) *to what extent does viscosity modify the dispersion and dissipation of deep-water surface waves propagating on a sheared background current?*

We limit our model to two-dimensional linear deep-water WCI at the ocean scale. By investigating these questions, we are addressing the concern voiced by Ellingsen and Brevik, and Dias et al.: that foundational results in this area are often inherited or cited rather than derived in general form from first principles. To the best of our knowledge, no comparable existing model derived from first principles combines linear shear current and viscous dissipation under ocean conditions. A similar gap appears to exist for models that explicitly incorporate a depth-dependent eddy viscosity. This thesis aims to close these gaps.

1.4 Project description

Chapter 2: Presentation of benchmark linear wave theory, including linear shear current, viscous boundary conditions and the standard result for deep water viscous dissipation rate. Uniform and depth-dependent eddy-viscosity parameterization are described.

Chapter 3: The two-dimensional viscous Navier-Stokes equation for a mean current with surface wave perturbations is linearized and reduced to only its WCI-coupled terms. The modification to the viscous terms when including a depth-dependent eddy viscosity is shown. A WKB expansion setup for the reduced equations is presented.

Chapter 4: The leading-order system of equations of the WKB expansion is presented. The leading-order is validated against the theory in §2 and its compatibility with an eddy viscosity turbulent closure is shown. The leading-order system of equations is solved for the case of a depth-invariant current and a linear shear current. The dispersion is investigated with regards to frequency and alteration of wave dissipation. Finally the governing equations and boundary conditions for the second-leading order of the WKB expansion are presented as a contribution to future research on the topics discussed in this thesis.

Chapter 5: The most important results of the thesis are summarized.

CHAPTER

TWO

THEORY

2.1 Limitations of Physical Regimes

2.1.1 Ocean Waves

In this thesis we investigate ocean waves and their interaction with currents. Without further specification, the term "ocean waves" can be understood in many ways, much as "waves" is ambiguous until their characteristics are defined. We limit our attention to waves propagating on the surface of a body of water on the scale of a lake or a sea, conditions we hereafter refer to as ocean conditions. Following the definition of Holthuijsen (2007), we distinguish the observed *surface elevation* from a *wave*. In a time record of observed waves, the surface elevation, $\eta(t)$, is the instantaneous elevation of the sea surface relative to a reference level. A wave is the profile of the surface elevation between two successive downward zero-crossings, where the zero level is the mean elevation over some period of time. We define the wave height H as the difference between the lowest and highest extent of surface elevation within the wave, and the wave period T as the time between two downward zero-crossings (see Figure 2.1.1).

The term "ocean waves" can describe many wave types arising from a variety of physical phenomena, for instance surges generated by low atmospheric pressure at the centre of a large storm, or tsunamis caused by earthquakes. In this thesis, *ocean waves* refers specifically to wind-generated gravity waves.

Wind-generated gravity waves arise from exposure to sustained wind fields and typically have periods $T < 30\,\mathrm{s}$ and wavelengths $\lambda < 1500\,\mathrm{m}$ (Holthuijsen, 2007). Within the generation zone they are short-crested and irregular; upon leaving it, they emerge as *swell*, which is typically regular. As the name indicates, we assume gravity to be the dominant restoring force and neglect surface tension. This assumption holds for $T > 0.25\,\mathrm{s}$ and $\lambda > 0.1\,\mathrm{m}$; below these scales we enter the capillary wave regime (Holthuijsen, 2007).

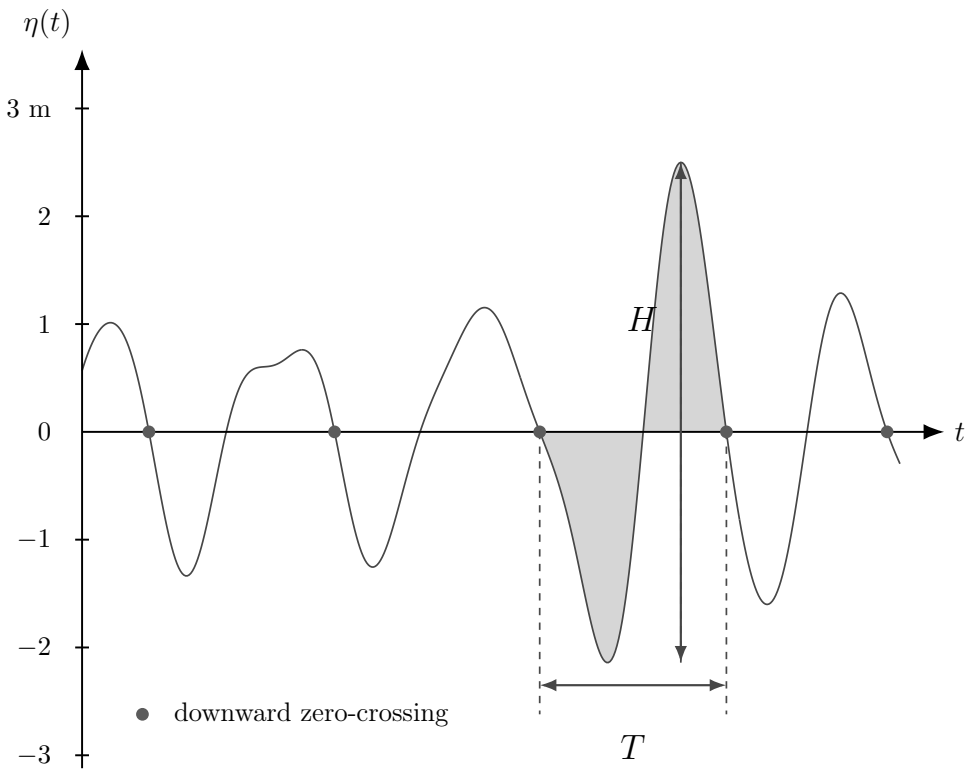


Figure 2.1.1: We define a wave as the profile between two subsequent downward zero-crossings. The shaded area constitutes a wave.

2.1.2 Ocean Currents

"Ocean currents" encompass a broad range of phenomena, from the basin-scale circulation of the great ocean gyres to the transient flows generated by local winds. In this thesis we are concerned with currents as background flow upon which the ocean waves propagate. We thus restrict the current to be horizontal mean flows, which change very slowly compared to the surface waves. Observed ocean currents are without exception turbulent (Cushman-Roisin & Beckers, 2011), which of course make them difficult to model exactly. We thus simplify the problem and restrict our attention to two current profiles: a depth-invariant current, and a linear shear current. These two cases are physically motivated as idealized models of tidal current, or ocean gyres (uniform in depth) and wind induced current.

Physically, tidal currents are large-scale waves driven by gravitational force from the moon and the sun acting on the entire water column. Tidal waves have periods of 24 and 12 hours (Pugh & Woodworth, 2014), so we can safely treat them as depth constant currents compared to ocean waves. The same holds for ocean gyres arising from large-scale pressure gradients and the Earth's rotation, such as the western boundary currents, examples of which are the Gulf Stream and the Agulhas Current.

We will model wind-induced currents as linear shear currents since they possess a vertical structure over the depth of wave-influence. When wind blows over the sea surface it exerts a stress that drives a turbulent, rotational current. Under steady, horizontally homogeneous conditions this current's magnitude is greatest

at the surface, and is essentially absent below a few tens of metres for typical wind speeds (Ardhuin et al., 2009).

2.2 Surface Waves

A well-known case of WCI, for which we have found analytical closed form solutions, is that of two-dimensional linear surface waves propagating on a linear shear current. This solution will be presented in the following subsections as it will serve as a benchmark for the results presented in § 4, in addition to describing many of the most important mechanisms in WCI. Building on basic theory, we will show that this particular problem also reproduces the general results of linear theory for waves only.

We define the example case as small waves propagating on the linear shear background current

$$U(z) = U_0 + U_b z, \tag{2.1}$$

where $U_b = U_0/h$ is the vorticity of the mean flow (Ellingsen & Brevik, 2014) and h is the depth of the system. We define the surface elevation as $\eta(x,t)$, and we assume it to be perturbed, thus oscillating freely with amplitude $A \ll h$. See Figure 2.2.1 for illustration.

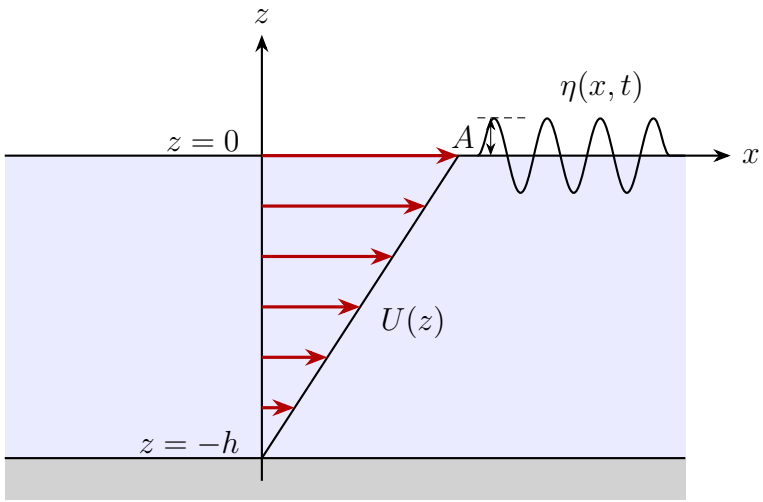


Figure 2.2.1: Linear shear current profile with small surface waves over finite depth.

2.2.1 Equations for incompressible flow

For the temporal and spatial scales of ocean conditions, the variation of water density is for most purposes insignificant (Mei et al., 2005). Under this assumption, water becomes incompressible and its motion is described by

- conservation of mass

$$\nabla \cdot \mathbf{u} = 0, \quad \text{and} \tag{2.2}$$

- conservation of momentum

$$\left(\frac{\partial}{\partial t} + \mathbf{u} \cdot \nabla\right) \mathbf{u} = -\nabla \left(\frac{P}{\rho} + gz\right) + \nu \nabla^2 \mathbf{u}, \tag{2.3}$$

where $\mathbf{u}(\mathbf{x}, t)$ is the flow velocity vector (u, w), $P(\mathbf{x}, t)$ the pressure, ρ the density, g the gravitational acceleration and ν the kinematic viscosity. $\mathbf{x}$ is the two-dimensional Euclidean space (x, z) where z points vertically upward. We call (2.2) the volume continuity equation, and (2.3) is the Navier–Stokes equation.

From (2.2) and (2.3) we can deduce the vorticity vector

$$\boldsymbol{\Omega} = \boldsymbol{\nabla} \times \mathbf{u}. \tag{2.4}$$

In fluid dynamics, a common class of problems is when the vorticity vector (2.4) equals zero. A flow which has zero vorticity is called an *irrotational flow* (Mei et al., 2005). In cases where we can also neglect the viscosity ν, we can simplify the description of the flow by expressing the velocity as the gradient of a scalar potential (Mei et al., 2005), i.e.

$$\mathbf{u}_{\text{irr}} = \nabla \Phi. \tag{2.5}$$

We call Φ the velocity potential. Using the velocity potential, the conservation of mass satisfies Laplace's equation,

$$\nabla^2 \Phi = 0. \tag{2.6}$$

The use of Φ generally simplifies fluid mechanical problems, and we wish to make use of its advantages whenever possible. It can be shown that a strictly two-dimensional flow with spatially constant vorticity $\boldsymbol{\Omega}_U = U_{b0}\hat{\mathbf{y}}$ at the initial time $t = 0$, will have unchanged vorticity for all time (Ellingsen & Brevik, 2014) (we have assumed the flow to live in the xz-plane). The WCI setup in Figure 2.2.1 satisfies these conditions, and we can therefore express the entire flow as a linear combination of the two dimensional linearly sheared mean flow and a perturbation in the form of a velocity potential, i.e.,

$$\mathbf{u} = U(z) + \nabla \Phi, \quad \text{where} \quad \nabla \times U(z) = U_b \hat{\mathbf{y}}. \tag{2.7}$$

2.2.2 Inviscid Boundary Conditions

We will derive the boundary equations for the flow described by (2.7). For ocean waves in a gravity field, two interfaces are of interest; the water-air interface at the upper boundary, and the water-solid interface at the bottom. Along both of these boundaries, we assume only tangential flow, ie. no water escapes either boundary.

2.2.2.1 The Kinematic Boundary Conditions

We will follow the derivation of Mei et al. (2005, p. 4) of the kinematic boundary condition at the surface and bottom. We express the equation for elevation of the surface boundary as

$$F(\mathbf{x}, t) = z - \eta(x, t) = 0, \tag{2.8}$$

where $\mathbf{x} = (x, z)$. x denotes the horizontal axis, z the vertical axis and η is the height measured from $z = 0$. (2.8) thus describes the surface boundary as free to move locally on the z-axis. We call such a moving surface a *free surface*. We denote a point $\boldsymbol{x}$ on the free surface as $\boldsymbol{q}$, so that when considering an infinitesimal time element dt, we can describe the surface as

$$F(\mathbf{x} + \mathbf{q}\,\mathrm{d}t,\, t + \mathrm{d}t) = 0 = F(\mathbf{x}, t) + \frac{\partial F}{\partial t}\,\mathrm{d}t + \mathbf{q} \cdot \nabla F\,\mathrm{d}t + \mathcal{O}(\mathrm{d}t^2) \tag{2.9}$$

We require of a fluid body that its surface is impermeable, i.e., that the particle on the surface must remain on the surface. We can thus only have tangential motion, so it follows from (2.9) that

$$\mathbf{u} \cdot \nabla F = \mathbf{q} \cdot \nabla F,$$

which again implies

$$\frac{\partial F}{\partial t} + \mathbf{u} \cdot \nabla F = 0, \quad \text{at } z = \eta. \tag{2.10}$$

We call (2.10), *the kinematic boundary condition*. We find the bottom boundary condition from the special case where equation (2.10) is the wetted surface of a stationary solid constraining the system at its bottom.

2.2.2.2 The Dynamic Surface Condition

On the free surface, both the elevation and the velocities are currently unknown, but we can study the forces governing the motion. For an inviscid system, the momentum equation reduces to the Euler equation

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla)\mathbf{u} = -\frac{1}{\rho}\nabla P - g\hat{\mathbf{z}}. \tag{2.11}$$

We use the vector identity $(\mathbf{u} \cdot \nabla)\mathbf{u} = \nabla\left(\frac{1}{2}|\mathbf{u}|^2\right) - \mathbf{u} \times \boldsymbol{\omega}$, so that (2.11) reads

$$\frac{\partial \mathbf{u}}{\partial t} + \nabla\left(\tfrac{1}{2}|\mathbf{u}|^2\right) - \mathbf{u} \times \boldsymbol{\Omega} = -\frac{1}{\rho}\nabla P - g\hat{\mathbf{z}} \tag{2.12}$$

With $\mathbf{u} = (U + \Phi_x)\hat{\mathbf{x}} + \Phi_z\hat{\mathbf{z}}$ and $\boldsymbol{\Omega} = U_b\hat{\mathbf{y}}$, the rotational term is

$$\mathbf{u} \times \mathbf{\Omega} = -U_b \frac{\partial \Phi}{\partial z}\,\hat{\mathbf{x}} + U_b\left(U(z) + \frac{\partial \Phi}{\partial x}\right)\hat{\mathbf{z}}. \tag{2.13}$$

We wish to express the Euler equation in terms of Φ, so we seek a scalar that satisfies

$$\mathbf{u} \times \mathbf{\Omega} = \nabla\chi = -U_b \frac{\partial \Phi}{\partial z}\,\hat{\mathbf{x}} + U_b\left(U(z) + \frac{\partial \Phi}{\partial x}\right)\hat{\mathbf{z}}. \tag{2.14}$$

Integrating the z-component and using that

$$\frac{\partial U}{\partial z} = U_b,$$

we find

$$\chi = \tfrac{1}{2}U(z)^2 + U_b \int^z \frac{\partial \Phi}{\partial x} dz' + C(x). \tag{2.15}$$

We demand that $\partial\chi/\partial x$ equals the x-component of (2.14), so

$$\frac{\partial \chi}{\partial x} = U_b \int^z \frac{\partial^2 \Phi}{\partial x^2} dz' + C'(x) = -U_b \frac{\partial \Phi}{\partial x}, \tag{2.16}$$

and use the fact that Φ satisfies Laplace's equation, i.e.,

$$\frac{\partial^2 \Phi}{\partial x^2} = -\frac{\partial^2 \Phi}{\partial z^2},$$

to see that

$$\int^z \frac{\partial \Phi}{\partial x} dz' + C_0 = -\int \frac{\partial \Phi}{\partial z} dx. \tag{2.17}$$

We can therefore rewrite (2.14) as

$$\mathbf{u} \times \mathbf{\Omega} = \nabla\left(\tfrac{1}{2}U(z)^2 - U_b \int \frac{\partial \Phi}{\partial z}\, dx\right). \tag{2.18}$$

Since the base flow is steady, $\partial\mathbf{u}/\partial t = \nabla(\partial\Phi/\partial t)$. Hence, upon substituting (2.18), (2.12) reduces to the gradient relation

$$\nabla\left(\frac{\partial \Phi}{\partial t} + \tfrac{1}{2}|\mathbf{u}|^2 - \tfrac{1}{2}U(z)^2 + U_b \int \frac{\partial \Phi}{\partial z}\, dx + \frac{P}{\rho} + gz\right) = 0, \tag{2.19}$$

which we integrate in space and find the unsteady Bernoulli equation for a flow of constant vorticity,

$$\frac{\partial \Phi}{\partial t} + \tfrac{1}{2}|U(z)\,\hat{\mathbf{x}} + \nabla\Phi|^2 - \tfrac{1}{2}U(z)^2 + U_b \int \frac{\partial \Phi}{\partial z}\,dx + \frac{P}{\rho} + gz = C(t), \tag{2.20}$$

The integration constant $C(t)$ may be absorbed into Φ and is hereafter set to zero.

Equation (2.20) is valid throughout the fluid domain. As we must have continuity of forces on the surface, the pressure just below the free surface must equal the pressure p_a above. We evaluate (2.20) on the free surface $z = \eta(x,t)$, expand the kinetic term and rearrange to find the full inviscid dynamic surface condition for a linearly sheared current,

$$-\frac{p_a}{\rho} = g\eta + \frac{\partial \Phi}{\partial t} + \big(U_0 + U_b\eta\big)\frac{\partial \Phi}{\partial x} + \tfrac{1}{2}|\nabla\Phi|^2 + U_b \int \frac{\partial \Phi}{\partial z}\,dx \quad \text{at } z = \eta. \tag{2.21}$$

2.2.2.3 Summarizing the Boundary Conditions

We invoke (2.7) and express the surface kinematic condition (2.10) and the bottom kinematic condition in terms of the velocity potential Φ. We summarize the inviscid boundary conditions for a two-dimensional linearly sheared flow at finite and constant depth:

- The kinematic bottom

$$\frac{\partial \Phi}{\partial z} = 0 \quad \text{at } z = -h, \tag{2.22}$$

- The kinematic surface

$$\frac{\partial \eta}{\partial t} + (U_0 + U_b\eta)\frac{\partial \eta}{\partial x} + \frac{\partial \eta}{\partial x}\frac{\partial \Phi}{\partial x} = \frac{\partial \Phi}{\partial z} \quad \text{at } z = \eta, \tag{2.23}$$

- The dynamic surface

$$g\eta + \frac{\partial \Phi}{\partial t} + \big(U_0 + U_b\eta\big)\frac{\partial \Phi}{\partial x} + \tfrac{1}{2}|\nabla\Phi|^2 + U_b \int \frac{\partial \Phi}{\partial z}\,dx = -\frac{p_a}{\rho} \quad \text{at } z = \eta, \tag{2.24}$$

where η is the elevation of the free surface, Φ is the velocity potential of the fluid and h is the depth of the system. ρ and g are respectively the density of the fluid and the gravitational constant.

2.2.3 Linearization

Throughout this thesis, we will use the linearized versions of the surface wave equations. Linearization enables us to consider surface waves and currents as independent entities while investigating their interactions as superpositions of their

initial properties. Linearized systems have certain limitations and are only valid for small surface elevations around the mean surface elevation, $z = 0$, and at low Reynolds numbers.

The assumptions for a linear surface wave-current system are that convective terms disappear and that the depth of the system varies slowly or remains constant. We make the following assumptions:

$$u \gg \max\{u^2,\, uv,\, uw\} \tag{2.25a}$$
$$\eta \gg \max\{\eta^2,\, u\eta\} \tag{2.25b}$$
$$|\nabla h| \ll 1 \tag{2.25c}$$

Under these assumptions, we find (2.6) and (2.22)-(2.24) reducing (in the same order) to

$$\nabla^2 \Phi = 0, \tag{2.26a}$$

$$\frac{\partial \Phi}{\partial z} = 0 \quad \text{at } z = -h, \tag{2.26b}$$

$$\frac{\partial \eta}{\partial t} + U_0 \frac{\partial \eta}{\partial x} = \frac{\partial \Phi}{\partial z} \quad \text{at } z = 0, \tag{2.26c}$$

$$\frac{\partial \Phi}{\partial t} + U_0 \frac{\partial \Phi}{\partial x} + U_b \int dx \, \frac{\partial \Phi}{\partial z} + g\eta = 0 \quad \text{at } z = 0. \tag{2.26d}$$

Note that we have set the reference level atmospheric pressure to zero in (2.26d). We assume the atmospheric pressure to be constant over the entire fluid domain, so the choice of p_a is a matter of gauge selection. We will assume $p_a = 0$ throughout the rest of this thesis.

2.2.4 The Dispersion Relation

To solve the system of equations (2.26), we follow Ellingsen and Brevik (2014). We assume

$$\Phi(x, z, t) = X(x)Z(z), \tag{2.27}$$

and see that to satisfy the governing equation (2.26a) and the bottom condition (2.26b), we must have a solution of the type

$$\Phi(x, z, t) = A(k, t)e^{ikx} \cosh k(z + h) \tag{2.28}$$

where $k = 2\pi/\lambda$, λ being the wavelength. An important insight is that the surface elevation must have the same wavelength as $\nabla\Phi$, so we write

$$\eta(x,t) = B(k,t)e^{ikx}. \tag{2.29}$$

Eliminating the coefficient $A(k,t)$ using (2.26c), we solve (2.26d) for $B(k,t)$ and its derivatives, and find the full solution for the motion of the surface elevation in a fixed frame of reference

$$\eta(x,t) = \beta_u(k)e^{i(kx+\sigma_u t - kU_0 t)} + \beta_d(k)e^{i(kx-\sigma_d t - kU_0 t)}, \tag{2.30}$$

where σ_u and σ_d are the angular frequencies in the frame of reference moving with the mean current. The subscripts u and d denotes the waves propagating upstream and downstream, respectively. We call σ_u and σ_d the intrinsic frequencies, and they demonstrate an important property of WCI; the Doppler shifted frequency. From the phase function of (2.30), we see it directly as

$$\sigma_{u,d} = \omega_{u,d} - kU(z=0). \tag{2.31}$$

The intrinsic frequency remains an important property throughout this thesis.

From (2.31) we find the fixed frame dispersion relations,

$$\omega_u(k) = \sqrt{gk\tanh kh + \left(\frac{U_b}{2}\tanh kh\right)^2} + \frac{U_b}{2}\tanh kh + kU_0, \tag{2.32a}$$

$$\omega_d(k) = \sqrt{gk\tanh kh + \left(\frac{U_b}{2}\tanh kh\right)^2} - \frac{U_b}{2}\tanh kh + kU_0, \tag{2.32b}$$

where $\omega_{u,d} = 2\pi/T_{u,d}$, $T_{u,d}$ being the wave period of the upstream and downstream propagating waves, respectively, measured in a fixed frame of reference.

2.2.5 The Deep Water Approximation

The solution in §2.2.4 is deduced under the assumption of finite depth. In this thesis, however, we are interested in ocean conditions where the depth usually is much larger than the wavelength of the waves. As a rule of thumb we can assume the depth to be infinite when $h > \lambda/2$ (Krogstad & Arntsen, 2000), thus relaxing the restrictions which the bottom boundary conditions lay on the system. However, if we naively let $h \to -\infty$ using the model deduced in the previous sections, (2.1) gives that $U_b \to 0$ and thus no shear. Throughout this thesis, we assume k to be real and positive and that that the current remains zero for depths beyond some limit $h*$, where $U(h*) = 0$.

2.2.6 Linear Benchmark Results

When assuming infinite depth we must use the asymptotic values for the linear dispersion relation. We let $h \to -\infty$ for which $\tanh kh \approx 1$. The dispersion relation (2.32) then reduces to

$$\omega_u(k) = \sqrt{gk + \frac{U_b^2}{4}} + \frac{U_b}{2} + kU_0, \tag{2.33a}$$

$$\omega_d(k) = \sqrt{gk + \frac{U_b^2}{4}} - \frac{U_b}{2} + kU_0, \tag{2.33b}$$

For the linear shear flow, we see that the dispersion differs for waves propagating upstream versus downstream and that the dispersion is Doppler shifted.

From (2.33), we will deduce two additional dispersions, depending on the properties of the flow. If we change the flow to being irrotational, i.e., setting $U_b = 0$, we get a depth-invariant current for which the dispersion simplifies to

$$\omega_u(k) = \omega_d(k) = \sqrt{gk} + kU_0. \tag{2.34}$$

We thus lose the difference in dispersion between the counter-propagating waves. However, the Doppler-shift remains. Finally, assuming no current,

$$\omega_u(k) = \omega_d(k) = \sqrt{gk}, \tag{2.35}$$

which is the well-known dispersion relation for linear surface waves under the deep water-approximation. We note that the presence of a mean current induces the Doppler shifted dispersion. The dispersion relations presented in this section will function as benchmarks and comparisons for the results found in §4.

2.3 Viscous Boundary Conditions

Even though useful for many large scale processes, the inviscid linear wave theory § 2.2 ignores important effects such as shear forces, wave damping, and the presence of turbulence. As we are developing a model aimed at fitting turbulence models, we must take viscosity into consideration. Viscosity is a fluid's resistance to shearing flows i.e., the friction between adjacent layers that move parallel to each other with different speeds. Viscosity thus gives rise to stresses in the interior of the fluid. Physically, stress is force per unit area, so for an infinitesimal surface area

$$\mathbf{A} = A_0\hat{\mathbf{n}},$$

we can define the stress vector,

$$\vec{f} = \boldsymbol{\sigma} \cdot \hat{\mathbf{n}} = \frac{\vec{F}}{A_0}, \tag{2.36}$$

where, in two dimensions and using the same conventions as before (Landau & Lifshitz, 1987),

$$\sigma_{ij} = -P\delta_{ij} + \rho\nu\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right), \quad i,j \in \{1,3\}. \tag{2.37}$$

P is the pressure of the fluid, and the second term is the viscous stress tensor. We can describe the motion of a viscous flow using the Navier-Stokes equation in terms of σ_{ij} (we remain in the linear regime) (Mei et al., 2005),

$$\frac{\partial u_i}{\partial t} = -g\delta_{i3} + \frac{1}{\rho}\frac{\partial \sigma_{ij}}{\partial x_j}. \tag{2.38}$$

Using (2.38), we will derive the linearized viscous boundary conditions. We assume deep water conditions, thus only requiring that the motion of the fluid dies out with depth, i.e.,

$$u_i \to 0, \quad \text{as } z \to -\infty.$$

At the surface, we still cannot have particle escape, a condition which is independent of the viscosity, so the kinematic conditions described in §2.2.3 remain unchanged. With viscosity, the dynamical surface condition now demands that we have continuity of stress through the free surface, i.e.,

$$\sigma_{ij}n_j = -\sigma_{ij}^{air}n_j, \quad \text{at } z = 0. \tag{2.39}$$

However, we assume (with good reason) that the air above the surface is inviscid, and thus cannot sustain stress. Using the justification from §2.2.3, we set the atmospheric pressure to zero, so the RHS of (2.39) vanishes.

Since the fluid is subject to the gravitational field, we can split the total pressure into a hydrostatic background and a dynamic perturbation

$$P = -\rho g z + \rho\frac{\partial \Phi}{\partial t} = -\rho g z + p.$$

We refer to p as the *modified pressure*: the flow-dependent component remaining once the hydrostatic contribution is removed.

With the right-hand side eliminated, the dynamic condition reduces to $\sigma_{ij}n_j = 0$ at $z = \eta$. For small-amplitude waves the outward unit normal is $\hat{\mathbf{n}} \approx (-\frac{\partial \eta}{\partial x}, 1)$ to first order. Retaining only linear terms of the stress balance (2.39), the normal ($i = 3$) and tangential ($i = 1$) components yield

$$p = \rho g\eta + 2\nu\rho\frac{\partial w}{\partial z}, \tag{2.40a}$$

$$\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x} = 0. \tag{2.40b}$$

We summarize the linearized viscous boundary conditions for incompressible flow at infinite depth (Mei et al., 2005).

- Deep-water condition:

$$u = w = 0, \quad \text{as } z \to -\infty, \tag{2.41}$$

- Kinematic surface condition (no current):

$$\frac{\partial \eta}{\partial t} = \frac{\partial \Phi}{\partial z}, \quad \text{at } z = 0, \tag{2.42}$$

- Kinematic surface condition (with current):

$$\frac{\partial \eta}{\partial t} + U_0 \frac{\partial \eta}{\partial x} = \frac{\partial \Phi}{\partial z}, \quad \text{at } z = 0, \tag{2.43}$$

- Dynamic surface conditions:

$$p = \rho g \eta + 2\nu\rho \frac{\partial w}{\partial z}, \quad \text{at } z = 0, \tag{2.44a}$$

$$\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x} = 0, \quad \text{at } z = 0. \tag{2.44b}$$

2.4 Viscous Decay Rate

The standard result for the viscous damping rate of surface gravity waves was first described by Lamb (1932). We will, however, present the derivation of Landau and Lifshitz (1987).

We assume that the viscosity of the water is small compared to the properties of the waves, i.e.,

$$\lambda^2 \omega \gg \nu, \quad A \ll \lambda,$$

A being the amplitude of the waves. This implies that the surface waves are small compared to the scales of the system, so the flow is only rotational in a relatively thin surface layer, while remaining ideal and irrotational throughout the rest of the fluid. We do, however, assume the velocity gradient to be much smaller than that of a viscous fluid-solid interface.

To calculate the energy dissipation, we must consider both the kinetic and potential energy of the waves. Again, we assume a two-dimensional system living in the xz-plane. Since gravitation does not affect the energy dissipation caused by internal friction (Landau & Lifshitz, 1987), we can describe the change of the total mechanical energy in the fluid as

$$\dot{E} = -\frac{1}{2}\rho\nu \int \left(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x}\right)^2 dV \tag{2.45}$$

where ν is the kinematic viscosity, u and w are, respectively, the horizontal and vertical velocity components of the wave motion and $\dot{E}$ is the derivative of the total energy with respect to time. Calculating the integral, we neglect the contributions from the irrotational surface layer since it is small compared to the rest of the system, and it does not carry a large velocity gradient. We can thus rewrite (2.45) using the velocity potential

$$\dot{E} = -2\rho\nu \int \left(\frac{\partial^2 \Phi}{\partial x \partial z}\right)^2 dV, \tag{2.46}$$

which we assume to take the generic form

$$\Phi = \phi_0 \cos(kx - \omega t + \alpha) e^{kz}.$$

We are interested in the mean values of the energy dissipation, so for the harmonic velocity potential we find

$$\overline{\dot{E}} = -8\rho\nu k^4 \int \overline{\Phi}^2 \, dV. \tag{2.47}$$

The total wave energy is the sum of kinetic and potential contributions. The mean kinetic energy density is $\frac{1}{2}\rho\,\overline{|\nabla\Phi|^2}$, so

$$\overline{E}_{\text{kin}} = \frac{1}{2}\rho \int \overline{|\nabla\Phi|^2}\, dV = \frac{1}{2}\rho \int \overline{\left(\frac{\partial \Phi}{\partial x}\right)^2 + \left(\frac{\partial \Phi}{\partial z}\right)^2}\, dV. \tag{2.48}$$

For Φ, both derivatives have equal mean square, $\overline{(\partial_x \Phi)^2} = \overline{(\partial_z \Phi)^2} = k^2\,\overline{\Phi^2}$, so

$$\overline{E}_{\text{kin}} = \rho k^2 \int \overline{\Phi^2}\, dV. \tag{2.49}$$

We invoke the virial theorem, which states that for small-amplitude oscillations the mean kinetic and potential energies are equal. The total mean energy is therefore twice the mean kinetic energy,

$$\overline{E} = 2\overline{E}_{\text{kin}} = 2\rho k^2 \int \overline{\Phi^2}\, dV. \tag{2.50}$$

For a damped harmonic oscillator, energy dissipates as

$$E(t) = E_0 e^{-2\gamma t}$$

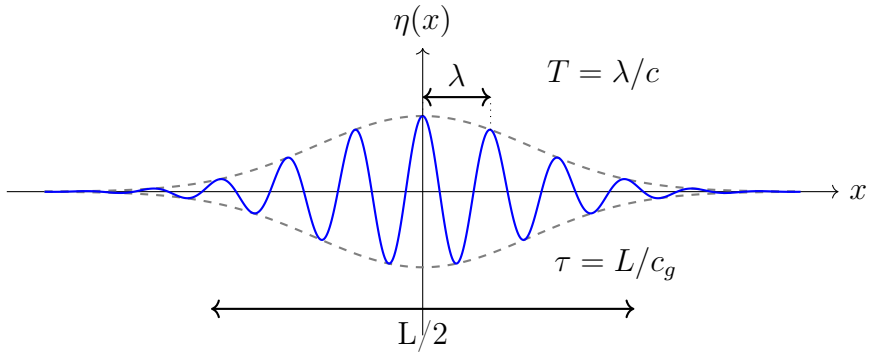


Figure 2.5.1: The surface elevation η in the form of a wave packet. The waves have wavelength λ and period T. The wave packet have wavelength L and period τ.

since it is proportional to the square of the wave amplitude. Solving for γ thus gives the damping coefficient of the latter,

$$\gamma = \frac{\overline{\dot{E}}}{2\overline{E}} = 2\nu k^2. \tag{2.51}$$

Note that for water, the kinematic viscosity $\nu \approx 1 \times 10^{-6}$, so for ocean waves we generally expect the damping rate to be small.

2.5 Perturbation Theory and Multiple-Scale Analysis

2.5.1 Perturbation Methods

Perturbation theory provides powerful tools for finding approximate solutions close to known, exact solutions. The idea is that in the vicinity of a solved problem, we can find good approximate solutions to related problems by expanding the exact solution as an asymptotic series of a small parameter μ:

$$A \equiv A_0 + \mu A_1 + \mu^2 A_2 + \mathcal{O}(\mu^3) \tag{2.52}$$

The first term of the series will then be the solvable part, while successive terms are small approximate variations to the known solution. By truncating the series we are left with a perturbation solution consisting of a known solution with a perturbation correction (Zheng & Zhang, 2017).

2.5.2 Multiple-Scale Analysis

To further discuss the theory of interactions between surface waves and currents, we need to introduce the mathematical concept of multiple-scale analysis. To de-

scribe multiple-scale analysis, we will first consider the wave packet in Figure 2.5.1. The rapidly oscillating wave has wavelength and period λ, T, while the envelope has the characteristic length and time scale L, τ. We assume that the properties of the rapidly oscillating wave is small, compared to those of the envelope, so that $\lambda \ll L$ and $T \ll \tau$. Secondly we assume that the two different length and time scales are independent of one another. Using these assumption, we can define a new set of coupled variables

$$\frac{x}{L} = \frac{\lambda}{L}\frac{x}{\lambda} = \mu\frac{x}{\lambda} \Rightarrow \bar{x}_L = \mu\bar{x}_\lambda, \tag{2.53}$$

$$\frac{t}{\tau} = \frac{T}{\tau}\frac{t}{T} = \mu\frac{t}{T} \Rightarrow \bar{t}_\tau = \mu\bar{t}_T, \tag{2.54}$$

where $\mu \ll 1$.

2.5.3 The WKB Method

The WKB method, named after Gregor Wentzel, Hendrik Anthony Kramers, and Léon Brillouin, is a special case of multiscale expansion used to approximate linear differential equations with spatially varying coefficients. It is useful in hydrodynamics because, rather than assuming an idealized potential flow carrying perturbative terms as described in § 2.5.1, it directly assumes a linear superposition of potential flows that vary on different scales (Li, 2017). Later in this thesis we use it to approximate solutions for interactions between surface waves and background currents.

The general idea of the WKB method (working in two-dimensional space) is to assume a wavefield of the form

$$\Psi(x,t) = \psi(x,t)\, e^{\frac{i}{\mu}S(x,t)}, \tag{2.55}$$

where $\psi(x,t)$ is the amplitude and $S(x,t)$ the phase function. As described in section 2.5.2, the parameter $\mu \ll 1$ is the ratio between the two time and spatial scales. The phase function $S(x,t)$ determines the local wavenumber and frequency of the wave, while the amplitude $\psi(x,t)$ varies slowly and governs the magnitude of the wavefield (Zwiebach, 2018).

To find approximate solutions for linear surface waves propagating over a slowly varying current or bottom topography, we substitute the WKB ansatz for $\Psi(x,t)$ into the governing linear differential equations and perform an asymptotic expansion. Each term in the expansion is organized according to its power of μ. We then solve the resulting hierarchy of equations in order of increasing power of μ.

Although the WKB method may, in principle, be carried out to any desired order, the assumption $\mu \ll 1$ means that the leading-order term $\mathcal{O}(1)$ dominates the next-order correction $\mathcal{O}(\mu)$, so it is typically sufficient to retain only the first two terms (Zwiebach, 2018).

2.6 Eddy Viscosity

At high Reynolds numbers, fluids becomes turbulent, resulting in chaotic mixing and enhanced transport of momentum. Just as molecular viscosity transfers momentum through molecular interactions, turbulence transports momentum down the gradient of the mean flow. This turbulent flux of momentum can be parameterized using an *eddy viscosity*, a phenomenological coefficient that plays a role analogous to molecular viscosity. Unlike molecular viscosity, which is a property of the fluid itself, eddy viscosity is a property of the flow. It varies with factors such as wave activity, shear, and stratification (Shen, 2019).

Observations constrain the magnitude of ν_e in the upper ocean. From ageostrophic current profiles in Drake Passage, Polton et al. (2013) showed that a constant, depth-uniform eddy viscosity of 0.08–$0.12\,\mathrm{m^2\,s^{-1}}$ reconciles the classical Ekman model with observations. Values of the same order are used in models by (Craik & Leibovich, 1976). Dias et al. (2008) recommend $\nu_e \sim 10^{-4}$–$10^{-3}\,\mathrm{m^2\,s^{-1}}$ for free-surface flows of waves only.

In the results presented in this thesis, we adopt the simplest closure of a uniform eddy viscosity and treat ν as a free parameter spanning $[10^{-6}, 10^{-2}]\,\mathrm{m^2\,s^{-1}}$, i.e., from the molecular viscosity of seawater and up to viscosities representative of a turbulent surface layer.

We aim to build a model capable of including more complex eddy-viscosity closures. An immediate advancement of the uniform parameterization is the depth-dependent eddy viscosity, (Durbin & Yang, 1992):

$$\frac{\partial}{\partial x_j}\left(\nu_e \frac{\partial u_i}{\partial x_j}\right), \tag{2.56}$$

where u_i is the i-th component of the velocity field (u, w). We will use (2.56) to verify that our model can handle viscosity profiles beyond a uniform closure.

CHAPTER

THREE

METHODS

3.1 The Governing Equations and Boundary Conditions

In this chapter, we derive the analytical framework on which the later investigation will be founded. We present a WKB expansion setup of a two-dimensional linear wave-current system, including viscous effects. We assume infinite water depth.

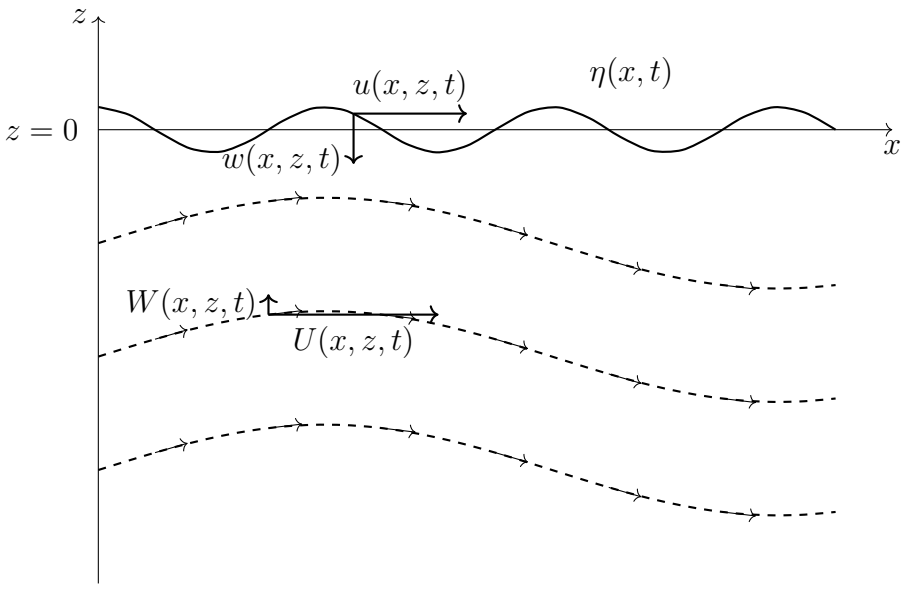


Figure 3.1.1: Linear surface wave with velocities $u(x,z,t), w(x,z,t)$, surface elevation $\eta(x,t)$ over a generalized slowly varying background current with components $U(x,z,t), W(x,z,t)$.

We consider a two-dimensional system consisting of water in motion, constrained by a horizontal free surface at $z = 0$ and infinite water depth $z \to -\infty$. The x-axis

lies in the horizontal plane and the z-axis in the vertical. See Figure 3.1.1 for configuration. The guiding equations of motion for this problem are the incompressible continuity equation (2.2) and the incompressible conservation of momentum (2.3). We separate the horizontal and vertical components of the incompressible momentum equation

$$\frac{\partial \tilde{u}}{\partial t} + \tilde{u}\frac{\partial \tilde{u}}{\partial x} + \tilde{w}\frac{\partial \tilde{u}}{\partial z} = -\frac{1}{\rho}\frac{\partial p}{\partial x} + \nu\left(\frac{\partial^2 \tilde{u}}{\partial x^2} + \frac{\partial^2 \tilde{u}}{\partial z^2}\right), \tag{3.1a}$$

$$\frac{\partial \tilde{w}}{\partial t} + \tilde{u}\frac{\partial \tilde{w}}{\partial x} + \tilde{w}\frac{\partial \tilde{w}}{\partial z} = -\frac{1}{\rho}\frac{\partial p}{\partial z} + \nu\left(\frac{\partial^2 \tilde{w}}{\partial x^2} + \frac{\partial^2 \tilde{w}}{\partial z^2}\right), \tag{3.1b}$$

where $\tilde{u}$ and $\tilde{w}$ are the horizontal and vertical velocities, ρ is the constant fluid density, p is the modified pressure, and ν is the viscosity of the fluid. The assumption of an incompressible fluid gives the volume continuity equation

$$\frac{\partial \tilde{u}}{\partial x} + \frac{\partial \tilde{w}}{\partial z} = 0. \tag{3.2}$$

We will evaluate our system under a set of linear boundary conditions. Due to the viscous terms in (3.1), we must use the viscous boundary conditions, described in §2.3, i.e.,

- continuity of tangential surface stress,

$$\frac{\partial \tilde{u}}{\partial z} + \frac{\partial \tilde{w}}{\partial x} = 0, \quad \text{at } z = 0, \tag{3.3}$$

- continuity of normal surface stress

$$p = \rho g \eta + 2\nu\rho\frac{\partial \tilde{w}}{\partial z}, \quad \text{at } z = 0, \tag{3.4}$$

- and the kinematic surface condition

$$\frac{\partial \eta}{\partial t} + \tilde{u}\frac{\partial \eta}{\partial x} = \tilde{w}, \quad \text{at } z = 0. \tag{3.5}$$

η is the surface elevation, and p is the modified pressure (2.3). The deep-water condition requires the velocities die out with depth, i.e.,

$$\tilde{u} = \tilde{w} = 0, \quad \text{as } z \to -\infty. \tag{3.6}$$

To investigate the interactions between the surface waves and the background current, we introduce the flow field $\tilde{u} = U + u$ and $\tilde{w} = W + w$, where U, W are the respective horizontal and vertical velocity components of the background current, and u, w the corresponding perturbative components of the free surface. We assume the current to vary slowly compared to the waves in time and the x-direction, but that they vary on the same scale in the z-direction.

We assume the waves to scale in the order of the wave steepness, $\mathcal{O}(kA)$, and the current as $\mathcal{O}(1)$ where $kA \ll 1$. As we are only interested in the interactions between the waves and the current, we evict from our equations, the terms which scales as $\mathcal{O}(1)$ and the nonlinear terms which scale as $\mathcal{O}(kA)^2$. We expand the equations of our combined wave-current system, (3.1)-(3.6), and show which terms are neglected, or kept for further discussion:

- The horizontal momentum equation

$$\begin{aligned}\overbrace{\cancel{\frac{\partial U}{\partial t}}}^{(*)} + \frac{\partial u}{\partial t} + \overbrace{U\cancel{\frac{\partial U}{\partial x}}}^{(*)} + U\frac{\partial u}{\partial x} + u\frac{\partial U}{\partial x} \\ + \overbrace{\cancel{u\frac{\partial u}{\partial x}}}^{\mathcal{O}(kA)^2} + \overbrace{\cancel{W\frac{\partial U}{\partial z}}}^{(*)} + W\frac{\partial u}{\partial z} + w\frac{\partial U}{\partial z} + \overbrace{\cancel{w\frac{\partial u}{\partial z}}}^{\mathcal{O}(kA)^2} \\ = -\frac{1}{\rho}\frac{\partial p}{\partial x} + \nu\overbrace{\left(\cancel{\frac{\partial^2 U}{\partial x^2}} + \cancel{\frac{\partial^2 U}{\partial z^2}}\right)}^{(*)} + \nu\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial z^2}\right). \quad (3.7)\end{aligned}$$

- The vertical momentum equation

$$\begin{aligned}\overbrace{\cancel{\frac{\partial W}{\partial t}}}^{=0} + \frac{\partial w}{\partial t} + \overbrace{U\cancel{\frac{\partial W}{\partial x}}}^{(*)} + U\frac{\partial w}{\partial x} + u\frac{\partial W}{\partial x} \\ + \overbrace{\cancel{u\frac{\partial w}{\partial x}}}^{\mathcal{O}(kA)^2} + \overbrace{\cancel{W\frac{\partial W}{\partial z}}}^{(*)} + W\frac{\partial w}{\partial z} + w\frac{\partial W}{\partial z} + \overbrace{\cancel{w\frac{\partial w}{\partial z}}}^{\mathcal{O}(kA)^2} \\ = -\frac{1}{\rho}\frac{\partial p}{\partial x} + \nu\overbrace{\left(\cancel{\frac{\partial^2 W}{\partial x^2}} + \cancel{\frac{\partial^2 W}{\partial z^2}}\right)}^{=0} + \nu\left(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial z^2}\right). \quad (3.8)\end{aligned}$$

- The incompressible continuity equation

$$\overbrace{\cancel{\frac{\partial U}{\partial x}}}^{(*)} + \overbrace{\cancel{\frac{\partial W}{\partial z}}}^{(*)} + \frac{\partial u}{\partial x} + \frac{\partial w}{\partial z} = 0. \quad (3.9)$$

- Continuity of tangential surface stress

$$\overbrace{\cancel{\frac{\partial U}{\partial z}}}^{(*)} + \overbrace{\cancel{\frac{\partial W}{\partial x}}}^{(*)} + \frac{\partial u}{\partial z} + \frac{\partial w}{\partial x} = 0, \quad \text{at } z = 0. \quad (3.10)$$

- Continuity of normal stress

$$p = \rho g\eta + 2\nu\rho\frac{\partial W}{\partial z} + 2\nu\rho\frac{\partial w}{\partial z}, \quad \text{at } z = 0. \quad (3.11)$$

- The kinematic surface condition

$$\frac{\partial \eta}{\partial t} + U\frac{\partial \eta}{\partial x} + \cancel{\overbrace{u\frac{\partial \eta}{\partial x}}^{\mathcal{O}(kA)^2}} = W + w, \quad \text{at } z = 0. \tag{3.12}$$

- The deep water condition

$$u \to 0, w \to 0 \quad \text{and} \cancel{\overbrace{U \to 0}^{(*)}}, \cancel{\overbrace{W \to 0}^{(*)}}, \quad \text{as } z \to -\infty. \tag{3.13}$$

$(*)$ Terms only depending on the background current.

We then collect the remaining WCI-terms from (3.7)-(3.13), which will form the basis of this investigation.

Governing Equations and Boundary Conditions

- **The horizontal momentum equation**

$$\frac{\partial u}{\partial t} + U\frac{\partial u}{\partial x} + u\frac{\partial U}{\partial x} + W\frac{\partial u}{\partial z} + w\frac{\partial U}{\partial z} + \frac{1}{\rho}\frac{\partial p}{\partial x} - \nu\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial z^2}\right) = 0 \tag{3.14a}$$

- **The vertical momentum equation**

$$\frac{\partial w}{\partial t} + U\frac{\partial w}{\partial x} + u\frac{\partial W}{\partial x} + W\frac{\partial w}{\partial z} + w\frac{\partial W}{\partial z} + \frac{1}{\rho}\frac{\partial p}{\partial z} - \nu\left(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial z^2}\right) = 0 \tag{3.14b}$$

- **The incompressible continuity equation**

$$\frac{\partial u}{\partial x} + \frac{\partial w}{\partial z} = 0 \tag{3.14c}$$

- **Continuity of tangential surface stress (at $z = 0$)**

$$\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x} = 0 \tag{3.14d}$$

- **Continuity of normal stress (at $z = 0$)**

$$p = \rho g \eta + 2\nu\rho\frac{\partial W}{\partial z} + 2\nu\rho\frac{\partial w}{\partial z} \tag{3.14e}$$

- **The kinematic surface condition (at $z = 0$)**

$$\frac{\partial \eta}{\partial t} + U\frac{\partial \eta}{\partial x} = W + w \tag{3.14f}$$

- **The deep water condition (as $z \to -\infty$)**

$$u \to 0, \quad w \to 0 \tag{3.14g}$$

3.2 Compatibility with Eddy Viscosity

A motivation for deriving (3.14) is to investigate the effect of eddy viscosity on surface wave propagation, so we first confirm that the governing equations accommodate it. We replace the uniform viscous terms in (3.14a) and (3.14b) with the depth-dependent eddy viscosity, (2.56), giving

$$\frac{\partial}{\partial x}\left(\nu_e(z)\frac{\partial u}{\partial x}\right)+\frac{\partial}{\partial z}\left(\nu_e(z)\frac{\partial u}{\partial z}\right), \quad \frac{\partial}{\partial x}\left(\nu_e(z)\frac{\partial w}{\partial x}\right)+\frac{\partial}{\partial z}\left(\nu_e(z)\frac{\partial w}{\partial z}\right). \tag{3.15}$$

Expanding the derivatives, we obtain

$$\nu_e(z)\frac{\partial^2 u}{\partial x^2}+\frac{\partial \nu_e(z)}{\partial z}\frac{\partial u}{\partial z}+\nu_e(z)\frac{\partial^2 u}{\partial z^2}, \tag{3.16}$$

$$\nu_e(z)\frac{\partial^2 w}{\partial x^2}+\frac{\partial \nu_e(z)}{\partial z}\frac{\partial w}{\partial z}+\nu_e(z)\frac{\partial^2 w}{\partial z^2}. \tag{3.17}$$

The model thus admits a depth-dependent eddy viscosity. We adopt uniform viscosity throughout this thesis and return to the general case (3.16)–(3.17) only as a direction for future work.

3.3 Approximation Scheme

Finding exact solutions to (3.14) would be a formidable task, as there are no known exact solution to the Navier-Stokes equation. We therefore seek approximate solutions using the methodological framework presented in §2.5, and expand (3.14) using the WKB method.

Our assumption is that the current $\Psi(x,z,t) = U(x,z,t) + W(x,z,t)$ is slowly varying on the length scales L in the x-direction and τ in time, while the oscillations on the free surface are small and rapidly varying. Both the waves and the current is dependent of the scale $h*$ in the z-direction for which we require that $|h|* > \lambda/2$ to remain in the deep water regime. We compare the scales of the wave current system,

$$(\omega\tau)^{-1} \sim (kL)^{-1} \sim \frac{|h*|}{L} = \mathcal{O}(\mu) \ll 1. \tag{3.18}$$

Following the procedure described in §2.5, we define the slowly varying variables of our system in terms of the scaling parameter $\mu \ll 1$ and conduct an asymptotic expansion of the variables in (3.14) in terms of these variables.

WKB Expansion Setup

- **Scaled variables**

$$\bar{x} = \mu x, \quad \bar{t} = \mu t \tag{3.19a}$$

- **Background flow**

$$U = U(\bar{x}, z, \bar{t}), \quad W = W(\bar{x}, z, \bar{t}) \tag{3.19b}$$

- **Pressure ansatz**

$$p(x, z, t) = [p_0 + \mu p_1 + \cdots] \, e^{i\frac{S}{\mu}} \tag{3.19c}$$

- **Horizontal velocity ansatz**

$$u(x, z, t) = [u_0 + \mu u_1 + \cdots] \, e^{i\frac{S}{\mu}} \tag{3.19d}$$

- **Vertical velocity ansatz**

$$w(x, z, t) = [w_0 + \mu w_1 + \cdots] \, e^{i\frac{S}{\mu}} \tag{3.19e}$$

- **Free surface elevation ansatz**

$$\eta(x, z, t) = [\eta_0 + \mu \eta_1 + \cdots] \, e^{i\frac{S}{\mu}} \tag{3.19f}$$

Where,

$$S = k\bar{x} - \omega\bar{t},$$

$$p_i = p_i(\bar{x}, z, \bar{t}), \quad \eta_i = \eta_i(\bar{x}, z, \bar{t}),$$
$$u_i = u_i(\bar{x}, z, \bar{t}), \quad w_i = w_i(\bar{x}, z, \bar{t}),$$

From the expanded variables in (3.19), we define the relations for the the phase function, the derivatives with respect to the slowly varying variables and the condition for conservation of wave crests.

Relations for Phase Function and Derivatives

$$k \equiv \frac{\partial S}{\partial \bar{x}}, \qquad \omega \equiv -\frac{\partial S}{\partial \bar{t}}, \tag{3.20a}$$

$$\frac{\partial}{\partial x} \equiv \mu \frac{\partial}{\partial \bar{x}}, \qquad \frac{\partial}{\partial t} \equiv \mu \frac{\partial}{\partial \bar{t}}, \tag{3.20b}$$

$$\frac{\partial k}{\partial \bar{t}} + \frac{\partial \omega}{\partial \bar{x}} = 0 \tag{3.20c}$$

Having defined the scaled variables we investigate the relationship between U and W. In (3.14c) we have already required that the perturbative terms satisfy conservation of mass indepentently, so we require the same of the current,

$$\mu \frac{\partial U}{\partial \bar{x}} + \frac{\partial W}{\partial z} = 0. \tag{3.21}$$

Integrating with respect to z we find

$$W(\bar{x}, z, \bar{t}) = \mu \int \frac{\partial U}{\partial \bar{x}} dz. \tag{3.22}$$

Thus we see that W is of order $\mathcal{O}(\mu)$.

Having now defined the coupling between the slowly varying current and the surface waves we are ready to proceed with the WKB expansion for each equation in (3.19).

CHAPTER

FOUR

RESULTS & DISCUSSION

4.1 The Leading-Order Equations

We insert the expansions (3.19) into (3.14). After computing the derivatives and reorganizing the expressions, we collect the terms according to their order in μ. At leading order, $\mathcal{O}(1)$, the resulting system of PDEs reads

System of PDEs for WCI – Leading order

- **The horizontal momentum equation**

$$i\frac{k}{\rho}p_0 + w_0\frac{\partial U(z)}{\partial z} - \left[i(\omega - kU(z)) + \nu\left(\frac{\partial^2}{\partial z^2} - k^2\right)\right]u_0 = 0 \tag{4.1a}$$

- **The vertical momentum equation**

$$\frac{1}{\rho}\frac{\partial p_0}{\partial z} - \left[i(\omega - kU(z)) + \nu\left(\frac{\partial^2}{\partial z^2} - k^2\right)\right]w_0 = 0 \tag{4.1b}$$

- **The incompressible continuity equation**

$$iku_0 + \frac{\partial w_0}{\partial z} = 0 \tag{4.1c}$$

- **The kinematic surface condition (at $z = 0$)**

$$w_0 = -i(\omega - kU(0))\eta_0 \tag{4.1d}$$

- **Continuity of normal stress (at $z = 0$)**

$$p_0 = \rho g\eta_0 + 2\nu\rho\frac{\partial w_0}{\partial z} \tag{4.1e}$$

- **Continuity of tangential stress (at $z = 0$)**

$$ikw_0 + \frac{\partial u_0}{\partial z} = 0 \tag{4.1f}$$

- **The deep water condition (as $z \to -\infty$)**

$$u_0 \to 0, \quad w_0 \to 0 \tag{4.1g}$$

Note that we have introduced the shorthand $U(\bar{x}, z, \bar{t}) \sim U(z)$ since we have no explicit dependence on $\bar{x}$ or $\bar{t}$ in the leading-order system of equations.

4.2 Model Validation

4.2.1 Recovery of Inviscid Linear Wave Theory

Before proceeding with further manipulations of the leading-order system, we pause to verify that the assumptions underlying the governing equations and the WKB expansion are consistent with the theory in §2.2.

To this end, we let the viscosity $\nu \to 0$ and assume a linear shear current $U(z) = U_0 + U_b z$ in the leading-order system. We will show in turn that (i) the continuity equation becomes Laplace's equation for the perturbation potential, (ii) both leading-order momentum equations reduce independently to a single linearized Bernoulli relation, and (iii) the kinematic and dynamic surface conditions recover their inviscid counterparts.

Since $U(z)$ is taken to be linear in depth, the base flow carries uniform vorticity, and the perturbation field can be represented through a velocity potential Φ, as discussed in §2.2.1. Since the leading-order equations have no explicit dependence on x, t or $\bar{x}, \bar{t}$, we ignore the difference in scale for the sake of simplicity, and assume a perturbation potential of the form

$$\Phi(x, z, t) = \Phi(z) e^{i(kx - \omega t)}, \tag{4.2}$$

so that

$$u_0 = \frac{\partial \Phi}{\partial x}, \quad w_0 = \frac{\partial \Phi}{\partial z}, \quad ik\Phi = \frac{\partial \Phi}{\partial x}, \quad i\omega\Phi = -\frac{\partial \Phi}{\partial t}. \tag{4.3}$$

4.2.1.1 The Governing Equations

The leading-order continuity equation, (4.1c), does not involve viscosity. Rewriting it using (4.3),

$$\frac{\partial^2 \Phi}{\partial z^2} + \frac{\partial^2 \Phi}{\partial x^2} = 0, \tag{4.4}$$

we see that it satisfies Laplace's equation directly, in agreement with (2.26a).

We show that, in the inviscid limit, the leading-order momentum equations reproduce the linearized constant-vorticity Bernoulli relation. For the linear shear current $U(z) = U_0 + U_b z$ the constant vorticity $U_b = \partial U/\partial z$. Linearizing the unsteady Bernoulli equation (2.20) (dropping the nonlinear kinetic terms and absorbing the hydrostatic part into the modified pressure p, see §2.3) gives the relation

$$\frac{p}{\rho} = -\frac{\partial \Phi}{\partial t} - U(z)\frac{\partial \Phi}{\partial x} - \frac{\partial U}{\partial z}\int \frac{\partial \Phi}{\partial z}\, dx. \tag{4.5}$$

We must verify that the leading-order momentum equations (4.1a) and (4.1b) are consistent with (4.5).

Setting $\nu = 0$ in the horizontal momentum equation (4.1a) and solving algebraically for p_0,

$$p_0 = \frac{\rho}{ik}\left[i\big(\omega - kU(z)\big)\, u_0 - \frac{\partial U}{\partial z}\, w_0 \right]. \tag{4.6}$$

Writing the velocities through the potential and using the derivative identities (4.3), this becomes

$$\frac{p_0}{\rho} = -\frac{\partial \Phi}{\partial t} - U(z)\frac{\partial \Phi}{\partial x} - \frac{1}{ik}\frac{\partial U}{\partial z}\frac{\partial \Phi}{\partial z}. \tag{4.7}$$

Since $\partial_x \Phi = ik\Phi$, the operator $1/(ik)$ acts as integration in x so (4.7) reads

$$\frac{p_0}{\rho} = -\frac{\partial \Phi}{\partial t} - U(z)\frac{\partial \Phi}{\partial x} - \frac{\partial U}{\partial z}\int \partial_z \Phi\, dx, \tag{4.8}$$

which equals (4.5); hence, the horizontal momentum equation is consistent with the linearized Bernoulli relation.

It remains to confirm that the vertical momentum equation is satisfied by the same p_0. Setting $\nu = 0$ in (4.1b),

$$\frac{1}{\rho}\frac{\partial p_0}{\partial z} - i\big(\omega - kU(z)\big)\, w_0 = 0. \tag{4.9}$$

Differentiating (4.5) in z, using (4.3) and Laplace's equation (4.4), the left-hand side reduces to

$$\frac{1}{\rho}\frac{\partial p_0}{\partial z} - i\big(\omega - kU(z)\big)\,\frac{\partial \Phi}{\partial z} = 0, \tag{4.10}$$

which is satisfied identically. Both leading-order momentum equations are thus consistent with the inviscid Bernoulli relation (4.5), confirming agreement with the inviscid theory.

4.2.1.2 The Boundary Conditions

Setting $\nu = 0$, the kinematic surface condition, the continuity of normal stress, and the deep-water condition reduce respectively to

$$w_0 = -i(\omega - kU(0))\eta_0, \quad \text{at } z = 0, \tag{4.11a}$$

$$-g\eta_0 + \frac{1}{\rho}p_0 = 0, \quad \text{at } z = 0, \tag{4.11b}$$

$$w_0 = 0, \quad \text{as } z \to -\infty. \tag{4.11c}$$

The continuity of tangential stress (4.1f) is a purely viscous effect and has no inviscid analogue, so it need not be verified here. The deep-water condition (4.11c) matches (2.26b) in the limit $z \to -\infty$. The surface elevation η_0 shares the wavelength and wave period of the perturbation potential (see 2.2.4), so under the ansatz (4.2) the kinematic surface condition (4.11a) reads

$$\frac{\partial \eta_0}{\partial t} + U(0)\frac{\partial \eta_0}{\partial x} = \frac{\partial \Phi}{\partial z}, \quad \text{at } z = 0, \tag{4.12}$$

which coincides with (2.26c).

The continuity of normal stress requires closer attention. Substituting for p_0 in (4.11b) and using (4.6) together with the identities (4.3) gives

$$g\eta_0 + \frac{\partial \Phi}{\partial t} + U(0)\frac{\partial \Phi}{\partial x} + \frac{1}{ik}\frac{\partial U}{\partial z}\frac{\partial \Phi}{\partial z} = 0. \tag{4.13}$$

As with (4.7), we anti-differentiate the $1/ik$-term, giving

$$\frac{\partial \Phi}{\partial t} + U(0)\frac{\partial \Phi}{\partial x} + \frac{\partial U}{\partial z}\int \frac{\partial \Phi}{\partial z}\, dx + g\eta_0 = 0. \tag{4.14}$$

The leading-order continuity-of-normal-stress condition therefore matches (2.26d) in the inviscid limit.

4.2.2 Consistency with Eddy Viscosity

Having confirmed the inviscid limit in §4.2.1, we now verify that the leading order of the WKB expansion remains valid when the depth-dependent eddy viscosity introduced in §3.2 is used in place of uniform viscosity. From the velocity ansatz (3.19d)–(3.19e), the variables $\bar{x}, \bar{t}$ are scaled by μ while z remains of $\mathcal{O}(1)$. The eddy-viscous terms (3.16)–(3.17) act only through z-derivatives and the $\mathcal{O}(1)$ profile $\nu_e(z)$, so they do not disturb the ordering in μ, and the expansion remains valid.

The viscous terms in the leading-order momentum equations (4.1a) and (4.1b) are therefore modified as

$$-\nu\left(\frac{\partial^2}{\partial z^2}-k^2\right)u_0 \quad \Longrightarrow \quad -\left[\frac{\partial \nu_e(z)}{\partial z}\frac{\partial}{\partial z}+\nu_e(z)\frac{\partial^2}{\partial z^2}-\nu_e(z)k^2\right]u_0, \tag{4.15}$$

and similarly for w_0. The equation for continuity of normal stress, (4.1e) contains the viscosity as a plain coefficient, so the substitution replaces ν by its surface value $\nu_e(0)$ and introduces no derivative of the profile. The leading-order model is therefore consistent with a depth-dependent eddy viscosity.

4.3 Simplification of the system of PDEs

As we have now ensured that the WCI system is consistent with inviscid linear wave theory, we proceed in our efforts towards solving (4.1). In our considerations we assume the viscosity ν and the shear current $U(z)$ to be known variables. From the leading-order system we see that the Doppler-shifted intrinsic frequency appears as an inherent part of the equations. We therefore introduce

$$\sigma(z) = \omega - kU(z), \tag{4.16}$$

which allows us to discuss the physics of the intrinsic frame of reference and moreover simplifies the equations somewhat.

4.3.1 The Governing Equations

As PDEs are substantially harder to solve than ODEs, a natural approach is to reduce the system of equations to an ODE. From the incompressible continuity equation (4.1c), we see that

$$u_0 = \frac{i}{k}\frac{\partial w_0}{\partial z}, \tag{4.17}$$

which we substitute into the horizontal momentum equation and rearrange into the following expression for p_0:

$$p_0 = \frac{\rho}{k^2}\left\{ ik\frac{\partial U(z)}{\partial z}w_0 + (i\sigma(z) - \nu k^2)\frac{\partial w_0}{\partial z} + \nu\frac{\partial^3 w_0}{\partial z^3}\right\} \tag{4.18}$$

We then differentiate (4.18) with respect to z and substitute for $\frac{\partial p_0}{\partial z}$ in the vertical momentum equation (4.1b). After reorganizing the equation and collecting terms of the vertical velocity component with respect to their order of differentiation in z, we find a fourth order ODE, describing the vertical velocity component of the surface waves:

$$\left(ik^2\sigma(z) - \nu k^4 - ik\frac{\partial^2 U(z)}{\partial z^2}\right)w_0 + \left(2\nu k^2 - i\sigma(z)\right)\frac{\partial^2 w_0}{\partial z^2} - \nu\frac{\partial^4 w_0}{\partial z^4} = 0 \tag{4.19}$$

To quickly assert that no error was made in rewriting (4.19), we check whether the units are consistent throughout the equation. We see that

$$\left[\nu\frac{\partial^4 w_0}{\partial z^4}\right] = \left[ik^2\sigma(z)w_0\right] = \left[\nu k^4 w_0\right] = \left[ik\frac{\partial^2 U(z)}{\partial z^2}w_0\right]$$
$$= \left[2\nu k^2\frac{\partial^2 w_0}{\partial z^2}\right] = \left[i\sigma(z)\frac{\partial^2 w_0}{\partial z^2}\right] = m^{-1}s^{-2},$$

thus showing that the units are consistent.

We rearrange (4.19) by saying that (4.19) can be written in the operator form

$$\hat{L}w_0 = 0, \tag{4.20}$$

where

$$\hat{L} \equiv \left(ik^2\sigma(z) - \nu k^4 - ik\frac{\partial^2 U(z)}{\partial z^2} \right) + \left(2\nu k^2 - i\sigma(z) \right) D^2 - \nu D^4, \tag{4.21}$$

and

$$\hat{D} \equiv \frac{\partial}{\partial z}.$$

To keep track of the derivatives of w_0 and $U(z)$ we deliberately keep the second partial derivative of $U(z)$ unchanged. We factorize the operator $\hat{L}$,

$$\hat{L} = -\nu(\hat{D}^2 - k^2)^2 - i\sigma(z)(\hat{D}^2 - k^2) - ik\frac{\partial^2 U(z)}{\partial z^2}, \tag{4.22}$$

so (4.19) can be rewritten

$$-\nu(\hat{D}^2 - k^2)^2 w_0 - i\sigma(z)(\hat{D}^2 - k^2)w_0 - ik\frac{\partial^2 U(z)}{\partial z^2}w_0 = 0. \tag{4.23}$$

4.3.2 The Boundary Conditions

As we are now considering an ODE in terms of w_0 and its derivatives with respect to z, we must also rewrite the boundary conditions in a similar manner. It can be easily seen that the continuity of tangential surface stress (4.1f) and the deep-water conditions (4.1g) can be rewritten

$$k^2 w_0 + \frac{\partial^2 w_0}{\partial z^2} = 0 \quad (\text{at } z = 0), \tag{4.24}$$

$$w_0 = \frac{i}{k}\frac{\partial w_0}{\partial z} = 0 \quad (\text{at } z \to -\infty), \tag{4.25}$$

using the incompressible continuity equation (4.1c). Meanwhile, we see that the kinematic condition, (4.1d), and the continuity of normal stress, (4.1e), both depend on η_0 and p_0. We reduce (4.1e) of p_0 by inserting (4.18) and rearrange both conditions as equations for η_0. We then have

- the kinematic surface condition (at $z = 0$)

$$\eta_0 = i\frac{w_0}{\sigma(0)} \tag{4.26}$$

- and the continuity of normal stress (at $z = 0$)

$$\eta_0 = \frac{1}{gk^2}\left\{(i\sigma(0) - 3\nu k^2)\frac{\partial w_0}{\partial z} + ik\frac{\partial U(z)}{\partial z}w_0 + \nu\frac{\partial^3 w_0}{\partial z^3}\right\}. \tag{4.27}$$

We equal the two conditions (4.26), (4.27) and rearrange in terms of w_0 and its derivatives. The resulting surface boundary condition reads

$$i\left(\frac{gk^2}{\sigma(0)} - k\frac{\partial U(z)}{\partial z}\right)w_0 + \left(3\nu k^2 - i\sigma(0)\right)\frac{\partial w_0}{\partial z} - \nu\frac{\partial^3 w_0}{\partial z^3} = 0. \tag{4.28}$$

We will from now on refer to this combined surface boundary condition as *the free surface condition*.

As a sanity check of the free surface condition we examine the units of each term of (4.28):

$$\left[\frac{gk^2}{\sigma}w_0\right] = \left[k\frac{\partial U(z)}{\partial z}w_0\right] = \left[3\nu k^2\frac{\partial w_0}{\partial z}\right] = \left[\sigma\frac{\partial w_0}{\partial z}\right] = \left[\nu\frac{\partial^3 w_0}{\partial z^3}\right] = s^{-2}.$$

which shows that the units are consistent throughout (4.28).

We have now reduced the leading-order system of PDEs, (4.1), to a fourth order ODE and four boundary conditions, all in terms of the vertical velocity component w_0 and its derivatives; thus constituting a solvable set.

Fourth order ODE and BCs for vertical velocity w_0

- **Governing ODE**

$$-\nu\left(\frac{\partial^2}{\partial z^2} - k^2\right)^2 w_0 - i\sigma(z)\left(\frac{\partial^2}{\partial z^2} - k^2\right)w_0 - ik\frac{\partial^2 U(z)}{\partial z^2}w_0 = 0 \tag{4.29a}$$

- **Free surface condition** (at $z = 0$)

$$i\left(\frac{gk^2}{\sigma(0)} - k\frac{\partial U(z)}{\partial z}\right)w_0 + \left(3\nu k^2 - i\sigma(z)\right)\frac{\partial w_0}{\partial z} - \nu\frac{\partial^3 w_0}{\partial z^3} = 0 \tag{4.29b}$$

- **Continuity of tangential stress** (at $z = 0$)

$$k^2 w_0(0) + \frac{\partial^2 w_0}{\partial z^2} = 0 \tag{4.29c}$$

- **Deep water-condition** (at $z \to -\infty$)

$$w_0 = \frac{i}{k}\frac{\partial w_0}{\partial z} \to 0 \tag{4.29d}$$

As indicated in §2.1, we are interested WCI under ocean conditions, and more precisely depth-invariant current and linear shear current. The second partial

derivative of the current is therefore not of interest. We set $\partial^2 U(z)/\partial z^2 = 0$, an assumption we will keep for all results presented in this section.

Assuming the current $U(z)$ to be linear or constant implies that

$$\sigma(z) \equiv \sigma_a + \sigma_b z, \tag{4.30}$$

where

$$\begin{aligned} \sigma_a &= \omega - kU_0, \\ \sigma_b &= -kU_b. \end{aligned}$$

We are thus left with the governing equation

$$\left(\frac{\partial^2}{\partial z^2} - k^2\right)^2 w_0 = -i\frac{(\sigma_a + \sigma_b z)}{\nu}\left(\frac{\partial^2}{\partial z^2} - k^2\right) w_0, \tag{4.31}$$

which is the Orr-Sommerfeld equation for a linear shear background current.

Key Result 4.1: The Orr-Sommerfeld equation

By reducing the WKB leading-order governing equations (4.1) to an ODE, we have recovered the Orr-Sommerfeld equation, which is much used in hydrodynamic stability analysis.

We will use (4.31) for two different case studies; i) a simplified model for ocean gyres or tidal current in the form of a depth invariant current, and ii) wind induced current modeled as a linear shear current, i.e.,

(i)
$$\sigma(z) = \sigma_a$$

(ii)
$$\sigma(z) = \sigma_a + \sigma_b z.$$

We will proceed by investigating the two cases separately.

4.4 Solution of the Leading-Order ODE with a Depth-Invariant Background Current

We will now proceed in our efforts towards finding a closed form solution to the case of a depth-invariant current. We set

$$U(z) = U_0, \quad \frac{\partial U(z)}{\partial z} = 0,$$

and

$$\sigma(z) = \sigma_a,$$

so that the governing equation and the free surface condition, respectively, reduce to

$$-\nu \left(\frac{\partial^2}{\partial z^2} - k^2 \right)^2 w_0 - i\sigma_a \left(\frac{\partial^2}{\partial z^2} - k^2 \right) w_0 = 0 \tag{4.32}$$

$$i\frac{gk^2}{\sigma_a} w_0 + \left(3\nu k^2 - i\sigma_a \right) \frac{\partial w_0}{\partial z} - \nu \frac{\partial^3 w_0}{\partial z^3} = 0. \tag{4.33}$$

The two-dimensional WCI system described in §3 is derived under linear assumptions, which dictates that the oscillations on the free surface are of a harmonic nature. The system in consideration also demands that the oscillations decay towards the bottom of the system. For such decaying linear harmonic oscillations, a sensible ansatz is that of a real exponential. As a solution to (4.32) we thus propose the ansatz

$$w_0(z) = Ae^{\lambda z}, \tag{4.34}$$

where A is some amplitude-related coefficient.

Upon inserting the ansatz into the governing equation, (4.29a), and dividing out w_0, we obtain a second degree polynomial in terms of λ^2, ie. a fourth degree polynomial in λ, yielding the roots

$$\lambda_{1\pm} = \pm k, \tag{4.35a}$$

$$\lambda_{2\pm} = \pm \left(k^2 - i\frac{\sigma_a}{\nu} \right)^{1/2}. \tag{4.35b}$$

4.4.1 The deep water boundary

We must now evaluate the roots (4.35) in the boundary condition for infinite depth. As previously discussed, a valid solution to our system must decay towards the

bottom of the system. Considering the exponential ansatz, we can thus directly rule out the roots that does not satisfy $\mathcal{R}e\{\lambda\} \geq 0$. In the case of (4.35a), it can easily be seen that we must discard the negative root of λ_1. As for (4.35b), we can define

$$\lambda_{2\pm} = \pm\left(k^2 - i\frac{\sigma_a}{\nu}\right)^{1/2} = \pm(u + iv),$$

where both u and v are real. Regardless of our choice of principle root, ie., whether we choose u to be positive or negative, the result is the same, and we must discard one of the two roots $\lambda_{2\pm}$. We are thus free to choose $u > 0$ so that only λ_{2+} satisfies the bottom boundary condition for $z \to \infty$.

The two valid roots are then

$$\lambda_1 \equiv \lambda_{1+} = k, \tag{4.36a}$$

$$\lambda_2 \equiv \lambda_{2+} = \left(k^2 - i\frac{\sigma_a}{\nu}\right)^{1/2}, \tag{4.36b}$$

giving a solution to $w_0(z)$ on the form

$$w_0(z) = A_1 e^{\lambda_1 z} + A_2 e^{\lambda_2 z}. \tag{4.37}$$

4.4.2 The surface boundary

We must evaluate (4.36) in the two surface boundary conditions, (4.33), (4.29c). It can easily be seen that upon inserting (4.37) into said equations, we get, for the free surface condition,

$$(\alpha + \beta\lambda_1 - \nu\lambda_1^3)A_1 + (\alpha + \beta\lambda_2 - \nu\lambda_2^3)A_2 = 0, \tag{4.38}$$

where,

$$\alpha = i\frac{gk^2}{\sigma_a} \tag{4.39}$$

$$\beta = 3\nu k^2 - i\sigma_a. \tag{4.40}$$

Meanwhile for the conserved tangential stress condition, we get

$$(k^2 + \lambda_1^2)A_1 + (k^2 + \lambda_2^2)A_2 = 0, \tag{4.41}$$

from which we directly read off the relation between the two amplitude coefficients A_1 and A_2:

$$\frac{A_1}{A_2} = -\frac{k^2+\lambda_2^2}{k^2+\lambda_1^2} = i\frac{\sigma_a}{2\nu k^2} - 1. \tag{4.42}$$

However, in accordance with the linear wave theory presented in §2.2, we cannot determine the amplitude of free surface oscillations from the leading-order alone. This thus remains a free parameter until having solved the second-leading-order.

The two surface conditions constitute a set of linear differential equations that has a nontrivial solution iff

$$\det\begin{pmatrix} \alpha+\beta\lambda_1-\nu\lambda_1^3 & \alpha+\beta\lambda_2-\nu\lambda_2^3 \\ k^2+\lambda_1^2 & k^2+\lambda_2^2 \end{pmatrix} = 0. \tag{4.43}$$

With some convenient algebra, we can simplify the determinant before proceeding towards finding a solution. We use that

$$\begin{aligned} \lambda_1 &= k, \\ \lambda_2^2 &= k^2 - i\sigma_a/\nu, \end{aligned}$$

and define

$$\mu \equiv -i\sigma_a/\nu = \lambda_2^2 - k^2,$$

to rewrite (4.43) as

$$\det\begin{pmatrix} \alpha+\nu k(2k^2+\mu) & \alpha+2\nu k^2\sqrt{k^2+\mu} \\ 2k^2 & 2k^2+\mu \end{pmatrix} = 0. \tag{4.44}$$

The determinant condition is thus satisfied when

$$\nu k\mu^2 + 4\nu k^3\mu + \left(4\nu k^5 + gk^2/\nu\right) = 4\nu k^4\sqrt{k^2+\mu}, \tag{4.45}$$

which upon substituting for μ gives

$$-\sigma_a^2 - 4i\nu k^2\sigma_a + 4\nu^2k^4 + gk = 4\nu^2k^3\sqrt{k^2 - i\sigma_a/\nu}. \tag{4.46}$$

Factorizing the first three terms and dividing through by ν^2k^4 gives us

$$\left(2 - \frac{i\sigma_a}{\nu k^2}\right)^2 + \frac{g}{\nu^2k^3} = 4\left(1 - \frac{i\sigma_a}{\nu k^2}\right)^{1/2}, \tag{4.47}$$

which is the relation of Dias et al. (2008) if we let $U_0 \to 0$ so that $\sigma_a \to \omega$.

Key Result 4.2: Dispersion relation for depth-invariant current

We have independently, via a WKB expansion of a full wave–current system, derived a dispersion relation of the same form as in Dias et al. (2008), but expressed in terms of the intrinsic frequency σ_a. The agreement with the classical viscous result validates our model as we have showed from first principles that a depth-invariant current enters only as a gauge transformation in the form of a Doppler shift. As seen in (2.34), the same holds for the inviscid dispersion relation.

Squaring both sides of (4.46), we find a quartic equation for σ_a:

$$\sigma_a^4 + 8ik^2\nu\sigma_a^3 - \left(2gk + 24k^4\nu^2\right)\sigma_a^2 - \left(8igk^3\nu + 16ik^6\nu^3\right)\sigma_a + g^2k^2 + 8gk^5\nu^2 = 0. \quad (4.48)$$

We conduct a simple sanity check on (4.48) by examining the units of each term:

$$\left[\sigma_a^4\right] = \left[8ik^2\nu\sigma_a^3\right] = \left[\left(2gk + 24k^4\nu^2\right)\sigma_a^2\right] = \left[\left(8igk^3\nu + 16ik^6\nu^3\right)\sigma_a\right] = \left[g^2k^2 + 8gk^5\nu^2\right] = s^{-4}.$$

We see that the units are consistent and in line with what we would expect for a quartic equation in terms of the intrinsic frequency σ_a.

(4.48) can be solved for σ_a to find an exact closed form dispersion relation. However, this expression is monstrous in its full size and will not allow for much analytical investigation of significance. We would also have to be aware of spurious roots introduced by the squaring of (4.45). The full closed form solution to the dispersion relation is, however, presented in Appendix A.

4.4.3 Validation and Physical Interpretation

4.4.3.1 Investigating Viscous Corrections to Inviscid Theory

Following Lamb (1932), Dias et al. (2008) expand (4.47) and truncate the expansion at $\mathcal{O}(\nu)$, which is well justified for molecular viscosity, where the relative size of the next-order term is below 10^{-3} for ocean wavelengths. They note, however, that practical applications call for eddy viscosities of order 10^{-4}–$10^{-3}\,\mathrm{m^2\,s^{-1}}$.

Based on the observations by Polton et al. (2013), we argue that the viscosity can be adjusted even higher in the presence of currents. We note, however, that a uniform eddy viscosity is the simplest turbulent-viscosity closure. It gives a useful basic description of the mean profile in simple shear flows but is incomplete and of limited applicability since the turbulent viscosity in reality varies significantly across the flow (Pope, 2000). We adopt it here as an idealization.

We will investigate if the Lamb dissipation rate is still a valid truncation of (4.46), if assuming that uniform eddy-viscosity parameterization can be used in the case of WCI. Upon increasing ν several orders of magnitude, the next order term is no longer guaranteed to be negligible relative to the Lamb dissipation rate. We therefore retain it and quantify its contribution *a posteriori*.

For $\nu \leq 10^{-2} m^2 s^{-1}$ and ocean wavelengths, $\nu k^2 \ll \sqrt{gk}$ still holds, so we take the angular frequency σ_a to consist of an inviscid base frequency and viscous correction terms. We can then expand the intrinsic angular frequency as the asymptotic series

$$\sigma_a = \sigma_{a0} + \sigma_{a1}\delta + \sigma_{a2}\delta^2 + \sigma_{a3}\delta^3 + \cdots , \tag{4.49}$$

where

$$\delta = \nu^{1/2}.$$

To make the order of the right-hand side of (4.46) explicit, we expand the expression under the root:

$$\begin{aligned} 4\nu^2 k^4 \sqrt{-i\sigma_a/\nu k^2} \cdot \sqrt{1 - \frac{\nu k^2}{i\sigma_a}} & \\ &\approx 4\nu^{3/2} k^3 \sqrt{-i\sigma_a} \left(1 - \frac{\nu k^2}{2i\sigma_a}\right) \\ &= \sqrt{-i\sigma_a} \left(4\nu^{3/2} k^3 - \frac{4\nu^{5/2} k^5}{2i\sigma_a}\right). \end{aligned} \tag{4.50}$$

Substituting the expanded right-hand side and (4.49) for σ_a into (4.46), and keeping terms up to third order in $\nu^{1/2}$, gives:

$$\begin{aligned} &- 2\left(2k^3\sqrt{-i\sigma_{a0}} + 2ik^2\sigma_{a1} + \sigma_{a1}\sigma_{a2} + \sigma_{a0}\sigma_{a3}\right)\delta^3 \\ &\quad + \left(-4ik^2\sigma_{a0} - \sigma_{a1}^2 - 2\sigma_{a0}\sigma_{a2}\right)\delta^2 - 2\delta\sigma_{a0}\sigma_{a1} + gk - \sigma_{a0}^2 + \mathcal{O}(\delta^4) = 0 \end{aligned} \tag{4.51}$$

Solving (4.51) at each leading order of δ gives:

$\mathcal{O}(\delta^0)$**:**

$$\sigma_{a0} = \sqrt{gk} \tag{4.52a}$$

$\mathcal{O}(\delta)$**:**

$$\sigma_{a1} = 0 \tag{4.52b}$$

$\mathcal{O}(\delta^2)$**:**

$$\sigma_{a2} = -2ik^2 \tag{4.52c}$$

$\mathcal{O}(\delta^3)$**:**

$$\sigma_{a3} = -\frac{2k^3}{\sigma_{a0}}\sqrt{-i\sigma_{a0}}. \tag{4.52d}$$

The branch of σ_{a3} is determined by the fact that we have already chosen right-propagating waves, i.e., $\sigma_{a0} = \sqrt{gk} > 0$, and that the root in (4.51) is the leading-order term of λ_2, for which we have already chosen $\mathrm{Re}(\lambda_2) > 0$. We must then have that

$$\sigma_{a3} = -2k^3\frac{\sqrt{-i}}{(gk)^{\frac{1}{4}}} = -2k^3\frac{e^{-i\frac{\pi}{4}}}{(gk)^{\frac{1}{4}}} = -\frac{\sqrt{2}\,k^3}{(gk)^{1/4}} + i\,\frac{\sqrt{2}\,k^3}{(gk)^{1/4}}. \tag{4.53}$$

The WKB leading-order dispersion relation is thus consistent with both inviscid linear wave theory (§2.2.4) and the viscous Lamb dissipation rate (§2.4) to the leading order in ν. We see that the next-order viscous correction lowers the real frequency of the wave field and weakens the dissipation rate. In the next section, we investigate the $\mathcal{O}(\delta^3)$ term.

We compare only the dissipation rates and omit a comparison of the real frequencies. For a depth-invariant current the fixed-frame frequency of both models is obtained by the identical Doppler shift, $\omega = kU + \sigma_a$, so any difference between the dispersion relations resides entirely in the intrinsic frequency, and the comparison reduces to the current-free case. There, the real part of the $\mathcal{O}(\nu^{3/2})$-correction is measured against the $\mathcal{O}(1)$ inviscid frequency $\sigma_{a0} = \sqrt{gk}$ and remains negligible even at the upper end of the eddy-viscosity range. A fixed-frame frequency comparison would thus add no information beyond the intrinsic one, and the practical impact of the $\mathcal{O}(\nu^{3/2})$-term is confined to the dissipation rate.

4.4.3.2 Comparing the Full Numerical Solution to the Expansions

We compare whether the viscous Lamb dispersion relation $\omega = \sqrt{gk} - 2i\nu k^2$ is a valid truncation of the WCI dispersion for a depth invariant current using uniform eddy-viscosity parameterization by comparing the deviation from a numerical root solution σ_{num} of the full dispersion relation (4.46). To assert that the numerical solver converge to the correct physical root, we assert for each solution that $\mathrm{Re}(\sigma_{\mathrm{num}}) > 0$, $\mathrm{Im}(\sigma_{\mathrm{num}}) < 0$ and that we have branch continuity for each solution (see Appendix B). We define the viscous dissipation rate terms from the expansion (4.52),

$$\gamma_2 = 2\nu k^2, \qquad \gamma_3 = -\frac{\sqrt{2}\,\nu^{3/2}k^3}{(gk)^{1/4}}, \qquad \gamma_{2,3} = \gamma_2 + \gamma_3,$$

and calculate the deviation from the full dispersion relation for a range of viscosity $\nu \in [10^{-6}\,\mathrm{m^2 s^{-1}}, 10^{-2}\,\mathrm{m^2 s^{-1}}]$ and wavelengths $\lambda \in [10\,\mathrm{m}, 300\,\mathrm{m}]$. The upper limit of 300m was chosen because, as we will see, we are to a large degree reproducing the inviscid results for wavelengths over 100m.

We observe the results in Table 4.4.1, and notice that for the eddy-viscosity ranges, $\nu_e \in [10^{-4}, 10^{-2}]$, the Lamb dissipation rate gives a deviation up of $0.022\% - 2.904\%$, while $\gamma_{2,3}$ yields a maximum deviation of 0.002%. We conclude that in the case of WCI, $\gamma_{2,3}$ improves the precision significantly compared to the Lamb dissipation rate and is thus a better analytical truncation in this regime, with the caveat that the eddy-viscosity results are indicative of the form and scaling of the correction rather than precise predictions.

Key Result 4.3: Significant $\mathcal{O}(\nu^{3/2})$ correction to dissipation rate for depth-invariant current

The classical dissipation rate of Lamb (1932), $\gamma_2 = 2\nu k^2$, remains accurate for molecular viscosity, but in the eddy-viscosity regime it deviates from the full dispersion relation by up to 3%. Retaining the $\mathcal{O}(\nu^{3/2})$ correction γ_3 reduces the maximum deviation to 0.002%. Since eddy-viscosity parameterizations are more relevant for WCI than for waves alone, the corrected rate $\gamma_{2,3}$ should be preferred in the WCI model.

ν [m^2/s]	Model	$\lambda = 10$ m	$\lambda = 30$ m	$\lambda = 100$ m	$\lambda = 300$ m
10^{-6}	γ_2	0.028%	0.012%	0.005%	0.002%
	$\gamma_2 + \gamma_3$	–	–	–	–
10^{-5}	γ_2	0.089%	0.039%	0.016%	0.007%
	$\gamma_2 + \gamma_3$	–	–	–	–
10^{-4}	γ_2	0.283%	0.124%	0.050%	0.022%
	$\gamma_2 + \gamma_3$	–	–	–	–
10^{-3}	γ_2	0.900%	0.393%	0.159%	0.070%
	$\gamma_2 + \gamma_3$	–	–	–	–
10^{-2}	γ_2	2.904%	1.253%	0.504%	0.221%
	$\gamma_2 + \gamma_3$	0.002%	–	–	–

Table 4.4.1: Percentwise deviation of the analytical solutions to the numerical root solution of the full viscous intrinsic dispersion relation (4.46). A dash indicates a deviation below 0.001%.

4.4.3.3 *e*-folding time corrections from the $\mathcal{O}(\nu^{3/2})$-term

Having established the analytic form of the higher-order viscous correction to the intrinsic frequency σ_a, we now quantify its practical impact on ocean wave dispersion and decay. We investigate the time a wave takes to be damped to e^{-1} of its original amplitude. We call this the *e*-folding time, $\tau = 1/\gamma$.

Figure 4.4.1 shows the absolute *e*-folding times for the γ_2-decay rate alone. For the molecular viscosity of water, the *e*-folding time exceeds seven days for all but the very shortest wavelengths and surpasses one year for wavelengths above 50 metres. Since the dissipation at molecular viscosity is already small at $\mathcal{O}(\nu)$, we expect the correction from our model to be very small in this range of ν. In the eddy-viscosity regime, however, substantial dissipation occurs within hours, so we expect a significant correction from the $\mathcal{O}(\nu^{3/2})$-term.

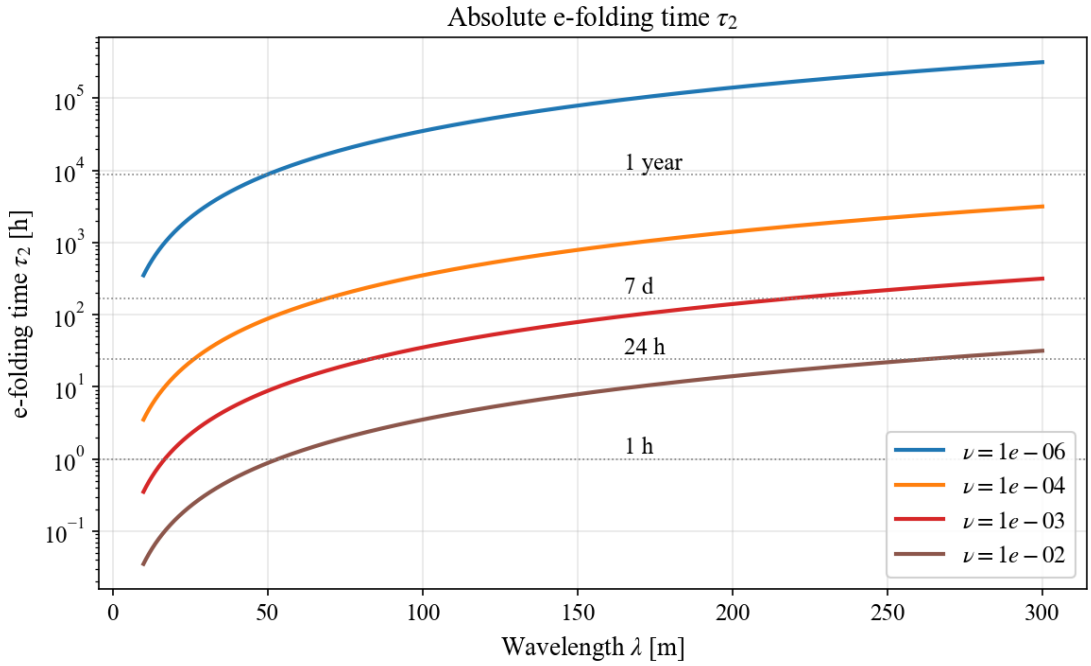


Figure 4.4.1: Absolute e-folding time $\tau_2 = 1/\gamma_2$ [h] from the $\mathcal{O}(\nu)$ dissipation rate, where $\gamma_2 = 2\nu k^2$. Horizontal reference lines mark the 1 h, 24 h, 7-day, and 1-year thresholds.

Figure 4.4.2 shows that the absolute correction to the e-folding time when including σ_{a3} is roughly six minutes to seven hours for molecular viscosity. This appears large but is negligible in relative terms, since the molecular e-folding times themselves span days to years, as Figure 4.4.3 confirms. In the eddy-viscosity regime the absolute correction is far smaller, ranging from a few seconds to about twelve minutes, yet because the underlying e-folding times are only hours long this represents a relative reduction in the dissipation rate of between one half and close to three percent. Whether these corrections are significant in applied settings lies beyond the scope of this analysis, as this would depend on the forcing and propagation history of the wave field, which is not modeled here.

4.4.4 Recovery of the Perturbation Amplitudes

At the beginning of this section, the leading-order system (4.1) was reduced to a single ODE in w_0. From its solution the full set of amplitudes $w_0(z)$, $u_0(z)$, and $p_0(z)$ can be reconstructed in closed form. These amplitudes are useful as input when advancing to the second-leading order of the WKB expansion; since that order is left to future work, they are reported in Appendix C rather than analyzed further here.

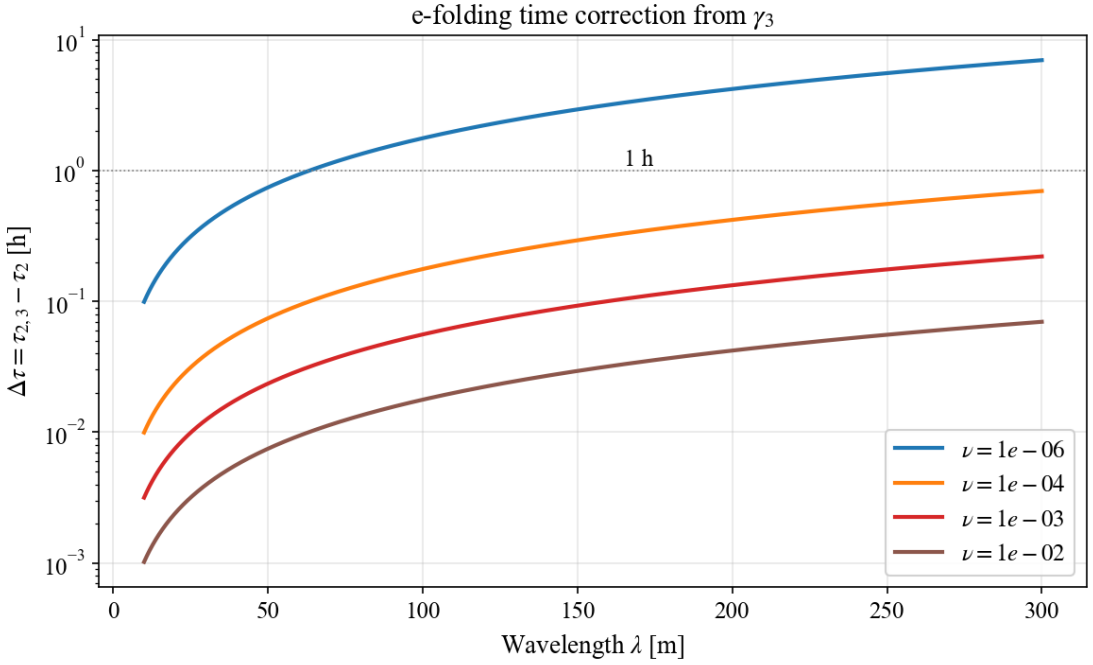


Figure 4.4.2: Absolute increase in e-folding time, $\Delta\tau = \tau_{2,3} - \tau_2$ [h], from the $\mathcal{O}(\nu^{3/2})$ correction, where $\tau_2 = 1/\gamma_2$ retains only the leading-order dissipation and $\tau_{2,3} = 1/(\gamma_2 + \gamma_3)$ includes both contributions. The horizontal reference line marks the 1 h threshold.

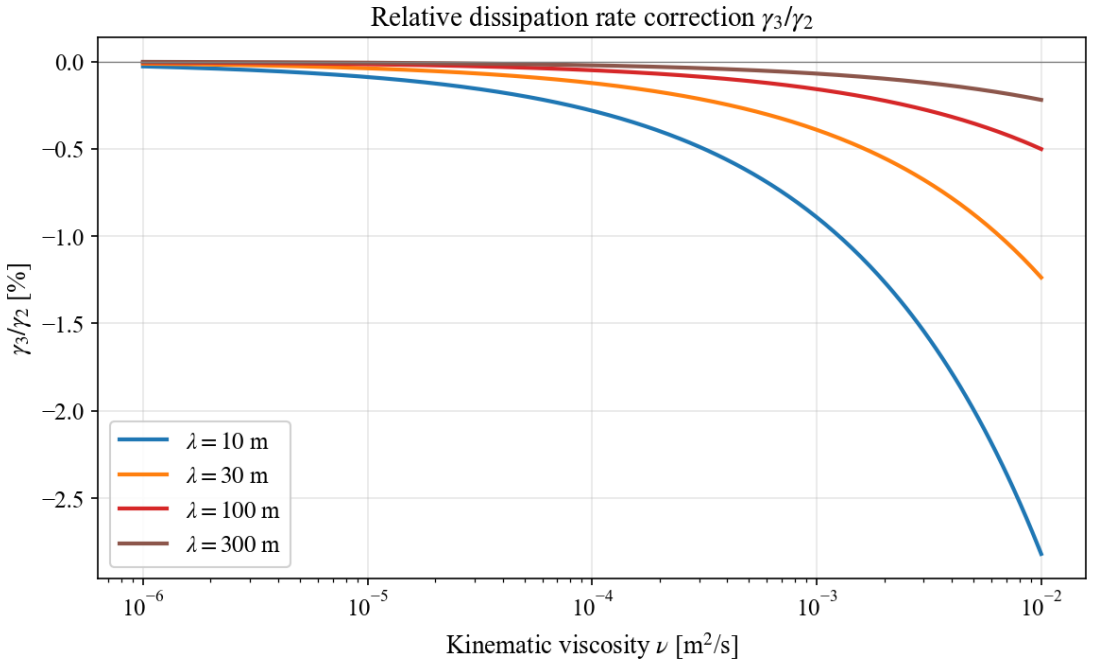


Figure 4.4.3: Relative dissipation rate correction γ_3/γ_2 [%], where $\gamma_2 = 2\nu k^2$ and $\gamma_3 = -\sqrt{2}\,\nu^{3/2}k^3/(gk)^{1/4}$.

4.5 Solution of the Leading Order ODE with a Linear Shear Current

We now consider the scenario in which the background current is linearly sheared along the vertical axis. As introduced at the beginning of this chapter, the intrinsic frequency then takes the form

$$\sigma(z) = \sigma_a + \sigma_b z.$$

We will roughly follow the analytical approach of (Feldman, 1957) and use $\sigma(z) = \omega - kU(z)$, where

$$U(z) = U_0 + U_b z,$$

so that the governing equation (4.31) becomes

$$\left(\frac{\partial^2}{\partial z^2} - k^2\right)^2 w_0 = -\frac{i(\omega - k(U_0 + U_b z))}{\nu}\left(\frac{\partial^2}{\partial z^2} - k^2\right) w_0. \tag{4.54}$$

Introducing the shorthand

$$R = \frac{U_b}{\nu}, \qquad c = \frac{\omega/k - U_0}{U_b},$$

equation (4.54) simplifies to

$$\left(\frac{\partial^2}{\partial z^2} - k^2\right)^2 w_0 = ikR(z - c)\left(\frac{\partial^2}{\partial z^2} - k^2\right) w_0. \tag{4.55}$$

To reduce the order of this equation, we apply the transformation introduced by Orr and Sommerfeld (Feldman, 1957). Setting

$$\frac{\partial^2 w_0(z)}{\partial z^2} - k^2 w_0(z) = \xi(z), \tag{4.56}$$

and substituting into (4.55) yields the second-order equation

$$\xi''(z) - \left(ikR(z - c) + k^2\right)\xi(z) = 0. \tag{4.57}$$

we see that by a change of variable, we can rewrite (4.57) as

$$f''(y) - y\, f(y) = 0, \tag{4.58}$$

We thus seek a linear substitution $y = A + Bz$, so that $z = (y - A)/B$. By the chain rule,

$$\frac{d^2\xi}{dz^2} = B^2 \frac{d^2\xi}{dy^2},$$

and (4.57) becomes, after dividing through by B^2,

$$\xi''(y) - \left(\frac{k^2 - ikRc}{B^2} - \frac{ikRA}{B^3} + \frac{ikR}{B^3}\, y \right) \xi(y) = 0. \tag{4.59}$$

Comparing with (4.58), the coefficients must satisfy

$$\frac{ikR}{B^3} = 1 \quad \text{and} \quad \frac{k^2 - ikRc}{B^2} - \frac{ikRA}{B^3} = 0,$$

giving $B = (ikR)^{1/3}$ and $A = (k^2 - ikRc)/(ikR)^{2/3}$. The substitution is therefore

$$y = \frac{k^2 - ikRc + ikRz}{(ikR)^{2/3}} = (ikR)^{1/3} \left(z - c - \frac{ik}{R} \right), \tag{4.60}$$

which transforms (4.57) into the Airy equation (see Appendix D)

$$\xi''(y) - y\,\xi(y) = 0. \tag{4.61}$$

The second-order nature of the Airy equation requires two linearly independent solutions. Since y is dependent on $i^{1/3}$ we find the two independent solutions along the contours of

$$e^{i(\pi/6 + 2\pi n/3)}, \qquad n = 0, 1$$

giving

$$\begin{aligned} \xi(z) = C_1\xi_1(z) + C_2\xi_2(z) = C_1 \mathrm{Ai} \left[e^{i\pi/6} (kR)^{1/3} \left(z - c - \frac{ik}{R} \right) \right] \\ + C_2 \mathrm{Ai} \left[e^{i5\pi/6} (kR)^{1/3} \left(z - c - \frac{ik}{R} \right) \right], \end{aligned} \tag{4.62}$$

where the Airy function of the first kind, $\mathrm{Ai}(y)$ is given by the improper Riemann integral (Lakshminarayanan & Varadharajan, 2015)

$$\mathrm{Ai}(y) = \frac{1}{\pi} \int_0^\infty \cos\left(\frac{\tau^3}{3} + y\tau \right) d\tau. \tag{4.63}$$

To find the full solution, we must thus solve (4.56) for $\xi(z)$. The equation is the modified Helmholtz equation with the nonhomogenous forcing term $\xi(z)$. The

fourth order nature of our original equation (4.54) requires four linearly independent solutions. Taking advantage of the linearity of the equation, we first solve the homogenous part

$$\frac{\partial^2 w_0}{\partial z^2} - k^2 w_0 = 0 \tag{4.64}$$

which we are familiar with from inviscid wave theory (§2.2), giving the two inviscid eigenmodes

$$\lambda_{1,2}(z) = e^{\pm kz}. \tag{4.65}$$

The method of variation of parameters gives the particular solution to the nonhomogeneous part (Boyce et al., 2017)

$$\lambda_{3,4}(z) = \lambda_2(z) \int_{z_1}^{z} \frac{\xi_{1,2}(t)\lambda_1(t)}{W(t)}\, dt - \lambda_1(z) \int_{z_0}^{z} \frac{\xi_{1,2}(t)\lambda_2(t)}{W(t)}\, dt, \tag{4.66}$$

where $W(t)$ is the Wronskian of the inviscid solutions,

$$W(t) = \det \begin{pmatrix} \lambda_1(t) & \lambda_2(t) \\ \lambda_1'(t) & \lambda_2'(t) \end{pmatrix} = -2k, \tag{4.67}$$

and t is an integration variable, not to be confused with time. z_0 and z_1 are constants that must be chosen to ensure convergence of the integrals. Expanding (4.66) gives

$$\lambda_{3,4}(z) = \int_{z_0}^{z} \frac{e^{k(z-t)}}{2k} \xi_{1,2} dt - \int_{z_1}^{z} \frac{e^{-k(z-t)}}{2k} \xi_{1,2} dt. \tag{4.68}$$

Examining the decay rate of $\xi_{1,2}$,

$$\xi_{1,2}(t) \sim \frac{1}{2\sqrt{\pi}\, t^{1/4}} \exp\left(-\frac{2}{3} |t|^{3/2} \right), \tag{4.69}$$

we see that if z_0, z_1 are chosen right, it dominates the integrand since $\mathrm{Ai}(y)$ decays exponentially as $y \to +\infty$ but is oscillatory for $y < 0$ (see Appendix D). Convergence for (4.68) is thus guaranteed if $z_0 = z_1 \to \pm\infty$, where the sign of of the limit is dependent on the argument of the Airy function. We rewrite the viscous eigenvalues using the hyperbolic sine function

$$\lambda_{3,4}(z) = \int_{\pm\infty}^{z} \frac{e^{-k(z-t)} - e^{k(z-t)}}{2k} \xi_{1,2} dt = \int_{\pm\infty}^{z} \frac{\sinh k(z-t)}{k} \xi_{1,2} dt, \tag{4.70}$$

which we later will see simplifies the expressions for the boundary conditions.

Now, noting that the real parts of the prefactors

$$e^{i\pi/6} = \cos\tfrac{\pi}{6} + i\sin\tfrac{\pi}{6},$$
$$e^{i5\pi/6} = \cos\tfrac{5\pi}{6} + i\sin\tfrac{5\pi}{6},$$

are of opposite signs, the arguments

$$y_1(t) = e^{i\pi/6}(kR)^{1/3}\left(t - c - \frac{ik}{R}\right), \tag{4.71a}$$
$$y_2(t) = e^{i5\pi/6}(kR)^{1/3}\left(t - c - \frac{ik}{R}\right) \tag{4.71b}$$

satisfy the convergence condition on the complementary domains of integration; $t \in (-\infty, z]$ and $t \in [z, +\infty)$, where the assignment of solution to domain is determined by the signs of the parameter R. Thus, for a background current following the wave direction, i.e. positive R, we find the viscous eigenvalues of (4.55) to be

$$\lambda_3(z) = \frac{1}{k}\int_{\infty}^{z} \sinh[k(z-t)]\,\xi_1(t)\,dt$$
$$= \frac{1}{k}\int_{\infty}^{z} \sinh[k(z-t)]\,\mathrm{Ai}\left[e^{i\pi/6}(kR)^{1/3}\left(t - c - \frac{ik}{R}\right)\right]dt, \tag{4.72a}$$

$$\lambda_4(z) = \frac{1}{k}\int_{-\infty}^{z} \sinh[k(z-t)]\,\xi_2(t)\,dt$$
$$= \frac{1}{k}\int_{-\infty}^{z} \sinh[k(z-t)]\,\mathrm{Ai}\left[e^{i5\pi/6}(kR)^{1/3}\left(t - c - \frac{ik}{R}\right)\right]dt. \tag{4.72b}$$

For an opposing current, i.e., $R < 0$, the viscous eigenvalues will swap integration domains.

4.5.1 The deep water boundary

We now determine which solutions of (4.55) satisfy the deep water boundary condition (4.29d). As in Section 4.4, the deep-water approximation requires that all solutions decay with depth, i.e.

$$\lambda_i(z) \to 0 \qquad \text{as} \qquad z \to -\infty, \quad i = 1, 2, 3, 4,$$

which selects, from the two inviscid solutions, only

$$\lambda_1(z) = Ae^{kz}. \tag{4.73}$$

For the viscous solutions $\lambda_{3,4}$, the admissible integration domain depends on the direction and vertical structure of the current. We consider first a following current with shear decaying to zero with depth, so that $R > 0$. The domain spans from the free surface at $z = 0$ to the bottom at $z \to -\infty$. Evaluating (4.72) in this limit gives

$$\lambda_3(z \to -\infty) = -\frac{1}{2k} \int_{-\infty}^{\infty} e^{-k(z-t)}\big|_{z\to-\infty} \cdot \mathrm{Ai}\left[e^{i\pi/6}(kR)^{1/3}\left(t - c - \frac{ik}{R}\right)\right] dt, \quad (4.74a)$$

$$\lambda_4(z \to -\infty) = \frac{1}{2k} \int_{-\infty}^{-\infty} e^{-k(z-t)}\big|_{z\to-\infty} \cdot \mathrm{Ai}\left[e^{i5\pi/6}(kR)^{1/3}\left(t - c - \frac{ik}{R}\right)\right] dt. \quad (4.74b)$$

It is readily seen that λ_3 diverges in this limit, whereas λ_4 vanishes identically as the integration domain collapses to a point.

The no-slip condition further requires

$$\left.\frac{\partial \lambda_i}{\partial z}\right|_{z\to-\infty} = 0,$$

which is trivially satisfied by (4.73). That it is also satisfied by λ_4 follows from an application of the Leibniz rule:

$$\left.\frac{\partial \lambda_4}{\partial z}\right|_{z\to-\infty} = \sinh(0)\, \xi(z)|_{z\to-\infty} + \int_{-\infty}^{-\infty} e^{k(z-t)}\big|_{z\to-\infty} \cdot \mathrm{Ai}\left[e^{i5\pi/6}(kR)^{1/3}\left(t - c - \frac{ik}{R}\right)\right] dt = 0, \quad (4.75)$$

again showing that the integration domain collapses to a point.

An entirely analogous argument applies to an opposing current decaying with depth, for which $R < 0$; and it is the solution with the kernel ξ_1, rather than ξ_2, that constitutes the admissible viscous eigenmode.

4.5.2 The surface boundary

As we are only left with two eigenmodes for each model setup, we rename them

$$\lambda_{\mathrm{invisc}}(z) \equiv \lambda_1(z) = e^{kz},$$

$$\lambda_{\text{visc}}(z) \equiv \begin{cases} \lambda_3(z) = \dfrac{1}{k}\displaystyle\int_{-\infty}^{z} \sinh[k(z-t)]\,\xi_1(t,R)\,dt & \text{if } R<0 \\ \lambda_4(z) = \dfrac{1}{k}\displaystyle\int_{-\infty}^{z} \sinh[k(z-t)]\,\xi_2(t,R)\,dt & \text{if } R>0 \end{cases}$$

where we have defined λ_{visc} as one variable depending on the direction of the shear current.

We follow the same procedure as in §4.4 by evaluating the eigenmodes in the free surface equation (4.29b)

$$\begin{aligned}[\alpha\lambda_{\text{invisc}}(0) + \beta\lambda'_{\text{invisc}}(0) - \nu\lambda'''_{\text{invisc}}(0)]\,A_1 \\ + [\alpha\lambda_{\text{visc}}(0) + \beta\lambda'_{\text{visc}}(0) - \nu\lambda'''_{\text{visc}}(0)]\,A_2 = 0,\end{aligned} \quad (4.77)$$

but now the coefficients include the shear current,

$$\alpha = i\left(\frac{gk^2}{\sigma(z)} - k\frac{\partial U(z)}{\partial z}\right), \quad (4.78)$$

$$\beta = 3\nu k^2 - i\sigma(z), \quad (4.79)$$

with

$$\sigma(z) = \sigma_a + \sigma_b z = \omega - kU_0 - kU_b z.$$

The conserved tangential stress condition, however, is unchanged wrt. the coefficients:

$$\left[k^2\lambda_{\text{invisc}}(0) + \lambda''_{\text{invisc}}(0)\right]A_1 + \left[k^2\lambda_{\text{visc}}(0) + \lambda''_{\text{visc}}(0)\right]A_2 = 0. \quad (4.80)$$

We examine the derivatives of λ_{visc} using Leibniz integral rule:

$$\begin{aligned}\lambda'_{\text{visc}}(z=0) &= \sinh(0)\xi(z=0) \\ &\quad + \int_{-\infty}^{z=0} \cosh k((z=0)-t)\,\text{Ai}\left[e^{i5\pi/6}(kR)^{1/3}\left(t - c - \frac{ik}{R}\right)\right]dt \\ &= \int_{-\infty}^{0} \cosh(-kt)\,\text{Ai}\left[e^{i5\pi/6}(kR)^{1/3}\left(t - c - \frac{ik}{R}\right)\right]dt\end{aligned} \quad (4.81a)$$

$$\begin{aligned}\lambda''_{\text{visc}}(z=0) = \cosh(0)\xi(z=0) + k\int_{-\infty}^{z=0} \sinh k((z=0)-t) \\ \cdot\,\text{Ai}\left[e^{i5\pi/6}(kR)^{1/3}\left(t - c - \frac{ik}{R}\right)\right]dt = k^2\lambda_{\text{visc}}(0) + \xi(0)\end{aligned} \quad (4.81b)$$

$$\lambda'''_{\text{visc}}(z=0) = k^2\lambda'_{\text{visc}}(0) + \xi'(0). \tag{4.81c}$$

Where we have used a general $\xi(t)$, dependent on the shear direction, ie.

$$\xi(t) = \text{Ai}\left[e^{i(\pi/6+2\pi n/3)}(kR)^{1/3}\left(t - c - \frac{ik}{R}\right)\right], \quad \text{where} \quad \begin{cases} n=1 & \text{if } R<0 \\ n=0 & \text{if } R>0. \end{cases}$$

We rewrite the surface conditions, (4.77) and (4.80), in terms of the derivatives of the eigenmodes, effectively lowering the order of the equations:

$$\left[\alpha + \beta k - \nu k^3\right] A_1 + \left[\alpha\lambda_{\text{visc}}(0) + \beta\lambda'_{\text{visc}}(0) - \nu\left(k^2\lambda'_{\text{visc}}(0) + \xi'(0)\right)\right] A_2 = 0 \tag{4.82a}$$

$$2k^2 A_1 + \left[2k^2\lambda_{\text{visc}}(0) + \xi(0)\right] A_2 = 0 \tag{4.82b}$$

Again, we can directly read off the relation between the two amplitude coefficients A_1 and A_2 from (4.82b):

$$\frac{A_1}{A_2} = -\frac{2k^2\lambda_{\text{visc}}(0) + \xi(0)}{2k^2} \tag{4.83}$$

Finally, we find the dispersion relation by solving the determinant condition for the surface system of linearly differentiable equations (4.82), resulting in the equation

$$2k^2(\beta - \nu k^2)\left(k\lambda_{\text{visc}}(0) - \lambda'_{\text{visc}}(0)\right) + (\alpha + \beta k - \nu k^3)\xi(0) + 2\nu k^2\xi'(0) = 0. \tag{4.84}$$

Seeing that

$$\begin{aligned}\left(k\lambda_{\text{visc}}(0) - \lambda'_{\text{visc}}(0)\right) &= \int_{-\infty}^{0} \sinh(-kt)\,\text{Ai}\left[e^{i5\pi/6}(kR)^{1/3}\left(t - c - \frac{ik}{R}\right)\right] dt \\ &- \int_{-\infty}^{0} \cosh(-kt)\,\text{Ai}\left[e^{i5\pi/6}(kR)^{1/3}\left(t - c - \frac{ik}{R}\right)\right] dt = -\int_{-\infty}^{0} e^{kt}\xi(t)dt,\end{aligned} \tag{4.85}$$

the determinant condition reduces to

$$2k^2(\nu k^2 - \beta)\int_{-\infty}^{0} e^{kt}\xi(t)dt + (\alpha + \beta k - \nu k^3)\xi(0) + 2\nu k^2\xi'(0) = 0. \tag{4.86}$$

Key Result 4.4: Dispersion relation for a viscous linear shear current

Via a WKB expansion of the full wave–current system, we have derived a leading-order dispersion relation, (4.86), for viscous gravity waves on a linear shear current in deep water. The relation is closed in terms of the Airy function $\xi(z)$ and a single integral over the viscous eigenmode. It holds for both shear directions through the choice of n, and couples viscosity and shear. The viscous eigenmodes are equal to Miesen and Boersma (1995), but they use different boundary conditions to analyze the stability of a liquid film on a solid. To the best of our knowledge, this dispersion relation has not previously been derived in the published literature; it generalises the viscous, current-free result of Dias et al. (2008) and the inviscid linear-shear result of Ellingsen and Brevik (2014) (taken to deep water) to a single model.

4.5.3 Comparison to the Inviscid Limit

Similar to §4.4, we must check whether the dispersion relation (4.86) is consistent with the inviscid theory in §2.2. Since the integral term complicates the analytical treatment, we investigate each term numerically for a set of physically realistic ocean values. See Table 4.5.1 for values.

Parameter	Value
λ	$100\,\mathrm{m}$
k	$2\pi/\lambda$
U_0	$0.1\,\mathrm{ms^{-1}}$
U_b	$0.04\,\mathrm{s^{-1}}$
g	$9.81\,\mathrm{ms^{-2}}$
ω	$0.5\,\mathrm{s^{-1}}$

Table 4.5.1: Set of physically realistic ocean-wave values.

In Figure 4.5.1 we see that both the first and last term of (4.86) fall off extremely fast as $\nu \to 0$, so the second term dominates the equation. We are left with

$$(\alpha + \beta k)\xi(0) = 0,$$

which when expanded gives a second order equation for ω with the roots,

$$\omega = \pm\sqrt{gk + U_b^2/4} - \frac{U_b}{2} + kU_0. \tag{4.87}$$

We assume right-propagating waves, so we must choose the positive root. Thus, (4.87) is equal to (2.33), showing that (4.86), within a margin of error, is consistent with the inviscid theory for ocean values.

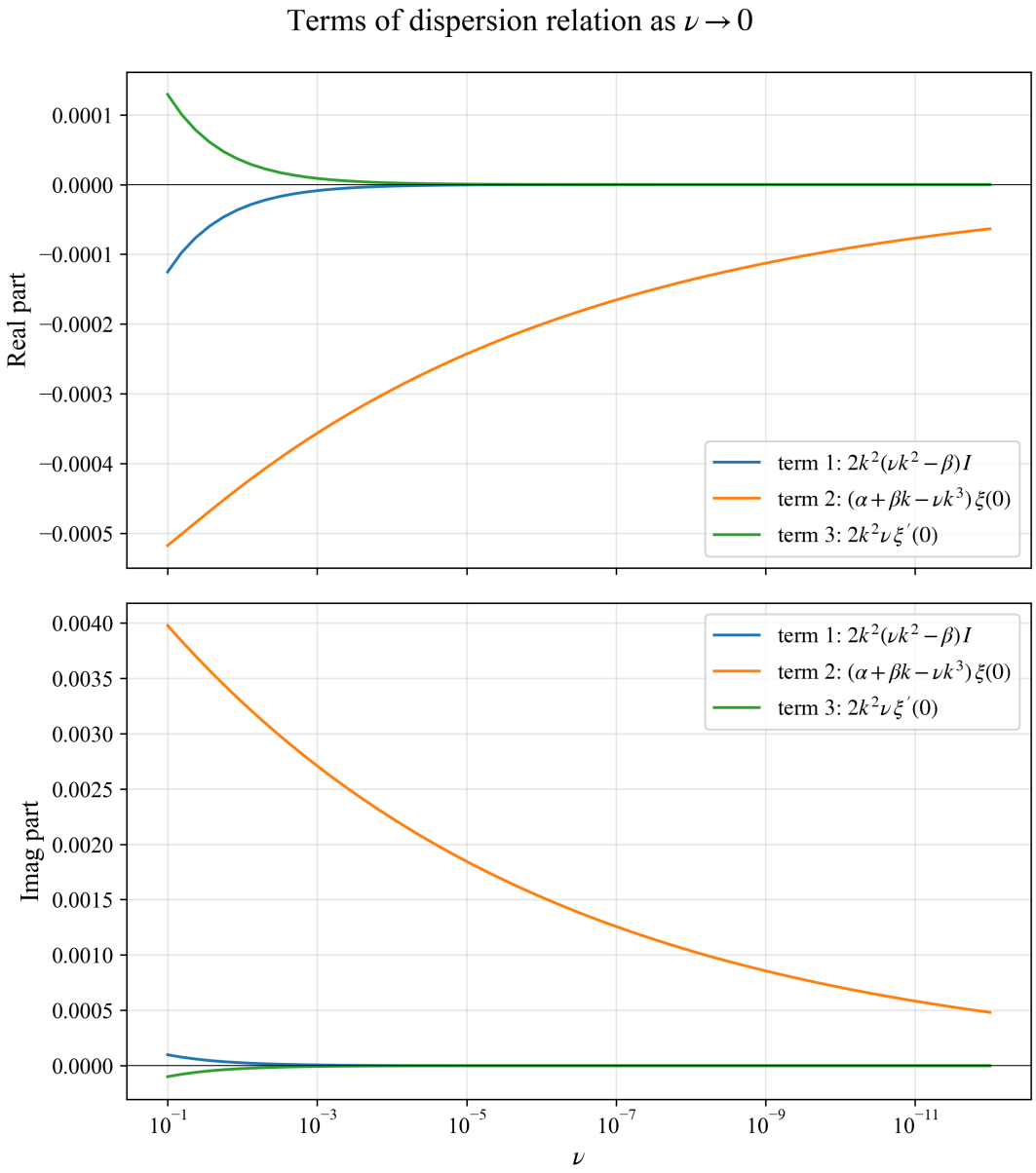


Figure 4.5.1: Real (upper) and imaginary (lower) parts of the three terms in the dispersion relation (4.86) as functions of viscosity ν, evaluated at the parameter values in Table 4.5.1. In term 1, $I = \int_{-\infty}^{0} e^{kt}\xi(t)dt$.

4.5.4 The Dispersion Relation for a Viscous Linear Shear Current

Now that we have shown consistency with the case of an inviscid linear shear current, we will proceed to analyze the dispersion relation (4.86). Our motivation for deriving this dispersion relation was to investigate idealized wind-induced WCI. We will therefore be considering parameters based on wind-ocean observations.

We rename the complex viscous dispersion

$$\omega \equiv \omega_{\text{visc}},$$

and solve (4.86) with respect to this. We invoke the benchmark results to which we will be comparing the viscous dispersion with shear. We will compare the wave frequency, $\text{Re}(\omega_{\text{visc}})$ to the inviscid dispersion relation we derived in §2.2.6

$$\omega_{\text{invisc}} = \sqrt{gk + U_b^2/4} - \frac{U_b}{2} + kU_0. \tag{4.88}$$

The dissipation rate $\text{Im}(\omega_{\text{visc}})$ will be compared against the dissipation rate we derived for depth-invariant current in §4.4,

$$\gamma_{U_0} \equiv \gamma_{2,3} = 2\nu k^2 - \frac{\sqrt{2}\,\nu^{3/2}k^3}{(gk)^{1/4}}. \tag{4.89}$$

Observations show that medium–strength winds induce ocean currents of between one to three percent of the wind speed at 10m above sea level (Holmedal & Myrhaug, 2013). We thus calculate ω_{visc} for a current of surface velocity $U_0 = 0.3\text{ms}^{-1}$ corresponding to a wind speed of $10\,\text{ms}^{-1}$, and shear $|U_b| \in [10^{-4}\text{s}^{-1}, |U_0|/(1.1 \times h*)]$, where $h* = \lambda/2$. Since we are working in the deep water approximation, we have restricted the maximum shears so that the current goes to zero outside the wave–influence depth with a ten percent margin for each wavelength. Letting the current change sign within this depth would have changed the physics of the system as we in that case would have superimposed a second current with a different forcing mechanism on the wind-induced shear current. We will investigate the viscosity–shear coupling for wavelengths in the span $\lambda \in [10\,\text{m}, 300\,\text{m}]$ (see Appendix E).

In Table 4.5.2 we see the maximum and minimum relative deviation between $\text{Re}(\omega_{\text{visc}})$ and ω_{invisc} over the shear span. Both the values for upstream and downstream propagating waves are tabulated. The correction is largest when the shear and viscosity have the highest values, and for waves propagating upstream i.e., for an upstream wavelength of ten meters and uniform eddy viscosity of $10^{-2}\,\text{m}^2\,\text{s}^{-1}$ we find a reduced frequency of $-9.96 \times 10^{-3}\%$. For a wavelength of 100 m, the maximum deviation is already in the order $\sim 10^{-5}\%$, so for practical applications in deep water ocean conditions, viscous wave frequency-correction to the case of an inviscid shear current is negligible. We notice that the viscous correction to the real frequency scales with ν and vanishes in the inviscid limit, consistent with the convergence shown in §4.5.3.

λ [m]	Direction	ν [m^2/s]	Min rel. dev [%]	Max rel. dev.[%]
10	downstream	1.0×10^{-6}	-8.34×10^{-9}	-8.56×10^{-9}
10	downstream	1.0×10^{-4}	-8.33×10^{-6}	-8.56×10^{-6}
10	downstream	1.0×10^{-2}	-8.33×10^{-3}	-8.54×10^{-3}
100	downstream	1.0×10^{-6}	-4.93×10^{-11}	-4.97×10^{-11}
100	downstream	1.0×10^{-4}	-4.93×10^{-8}	-4.97×10^{-8}
100	downstream	1.0×10^{-2}	-4.93×10^{-5}	-4.97×10^{-5}
300	downstream	1.0×10^{-6}	-4.20×10^{-12}	-4.22×10^{-12}
300	downstream	1.0×10^{-4}	-4.20×10^{-9}	-4.22×10^{-9}
300	downstream	1.0×10^{-2}	-4.20×10^{-6}	-4.22×10^{-6}
10	upstream	1.0×10^{-6}	-9.70×10^{-9}	-9.99×10^{-9}
10	upstream	1.0×10^{-4}	-9.70×10^{-6}	-9.99×10^{-6}
10	upstream	1.0×10^{-2}	-9.70×10^{-3}	-9.96×10^{-3}
100	upstream	1.0×10^{-6}	-5.17×10^{-11}	-5.21×10^{-11}
100	upstream	1.0×10^{-4}	-5.17×10^{-8}	-5.21×10^{-8}
100	upstream	1.0×10^{-2}	-5.17×10^{-5}	-5.21×10^{-5}
300	upstream	1.0×10^{-6}	-4.32×10^{-12}	-4.34×10^{-12}
300	upstream	1.0×10^{-4}	-4.32×10^{-9}	-4.34×10^{-9}
300	upstream	1.0×10^{-2}	-4.32×10^{-6}	-4.34×10^{-6}

Table 4.5.2: Maximum and minimum relative frequency deviation $(\mathrm{Re}(\omega_{\mathrm{visc}}) - \omega_{\mathrm{invisc}})/|\omega_{\mathrm{invisc}}|$ over the span of shears $|U_b| \in [10^{-4}\mathrm{s}^{-1}, |U_0|/(1.1 \times h*)]$ in percent. The maximum and minimum deviations correspond to the maximum and minimum shear for each wavelength.

Turning our attention to the dissipation rates, we have calculated the shear dissipation rate correction to that of the depth-invariant case. In Figure 4.5.2 and Figure 4.5.3 we have plotted the viscous dissipation rate corrections given by the shortest and longest investigated wavelengths. We see that the dissipation rate is equal for both current directions and that the presence of shear alters the dissipation rate only slightly compared to what was found in §4.4. The viscosity-shear coupling gives a correction to γ_{U_0} of $\sim -0.05\%$ which is about 25 times the magnitude of the 0.002% maximum deviation between γ_{U_0} and the full viscous dispersion relation (4.46) (see Table 4.4.1). Since the shear correction is clearly larger than the truncation error of γ_{U_0} it is evident that the shear physically changes the dissipation rate, but only marginally.

The correction $(|\operatorname{Im}(\omega_{\text{visc}})| - \gamma_{U_0})/\gamma_{U_0}$ reduces the dissipation rate in the eddy-viscosity regime, which increase the e-folding times of the waves. We see from Figure 4.5.4 and Figure 4.5.5 that the added e-folding times are in the order of seconds for most parameters investigated, just passing one minute for $\lambda = 300\,\text{m}$ at molecular viscosity. However, we remember from Figure 4.4.1 that for the latter, the absolute e-folding time was on the order of years. In other words, these corrections are negligible for swell under ocean conditions.

Due to negligible corrections to established theory, it can be discussed whether the viscous-shear dispersion relation (4.86) gets to couple the viscosity and shear before the $\text{Ai}(y(z))$ decays to zero as it has, under the assumptions used in this thesis, not been able to show any new effect of the viscous-shear coupling. Coupling effects may enter at the second-leading order as it is here that the multiple scales of the WKB expansion comes into play. We also ask ourselves whether a depth-dependent eddy viscosity with a much larger value close to the surface could compensate for the relatively weak wind-induced shears under ocean conditions, since it is only very close to the surface that the Airy function $\text{Ai}(y(z))$ is nonzero. We encourage further investigation to answer this question.

However (4.86) has shown strong consistency with both the known inviscid dispersion and viscous dissipation rates. This validate the viscous linear shear dispersion relation as a self-consistent model capable of unifying the standard analytical results for wave dispersion on a linear shear current with viscous dissipation. Recovery of all known limiting cases further supports the physical and mathematical foundations derived from first principles in §3, and builds confidence for the physical robustness of future work within this framework.

Key Result 4.5: Negligible shear corrections to the viscous linear-shear dispersion relation

When evaluated for realistic wind-induced current parameters, the dispersion relation (4.86) produces only small and well-behaved corrections to established theory. The real wave frequency deviates from inviscid shear theory by below $\sim 10^{-2}\%$, while the shear correction to γ_{U_0} reaches a maximum of $\sim 0.05\%$ and adds only seconds to e-folding times of order years. These physically consistent and negligible corrections validate the derived relation, although they reveal no new effect of the viscous-shear coupling. Such an effect may enter at the second-leading order, and we hypothesise that shear corrections may become important under more advanced eddy-viscosity closures.

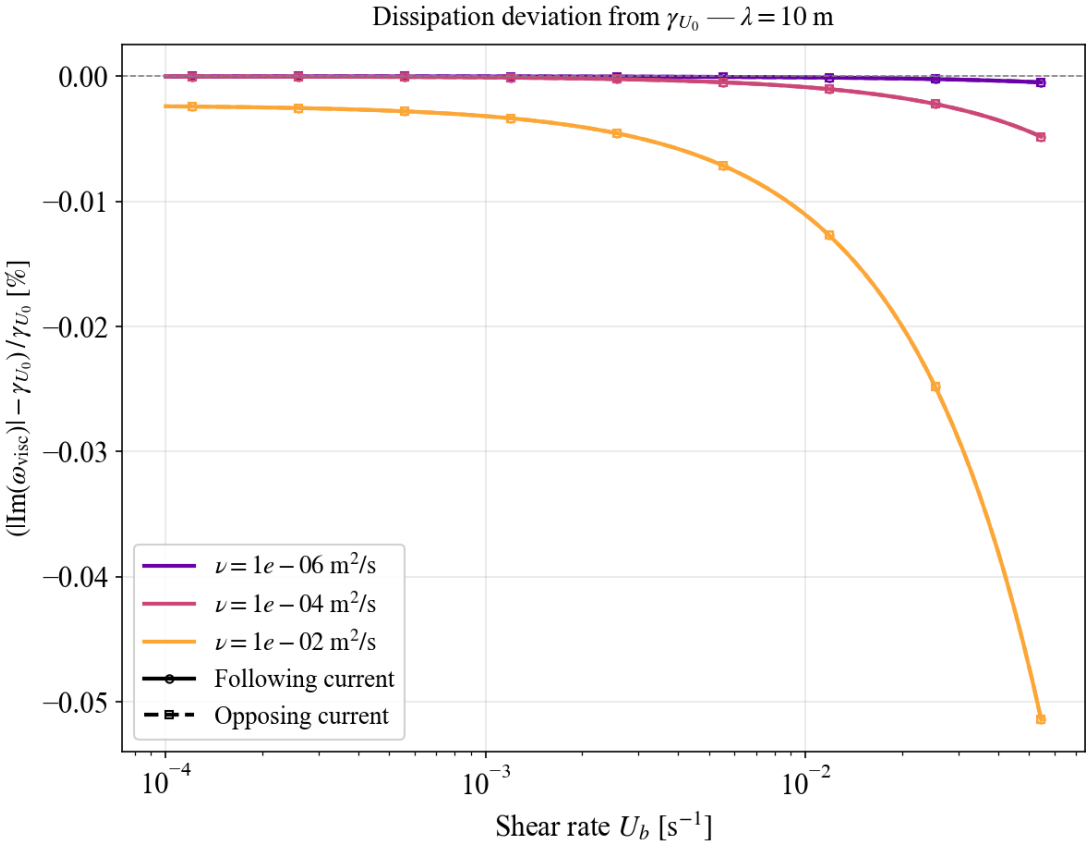


Figure 4.5.2: Relative dissipation-rate deviation for $\lambda = 10\,\mathrm{m}$, $(|\,\mathrm{Im}(\omega_{\mathrm{visc}})| - \gamma_{U_0})/\gamma_{U_0}$.

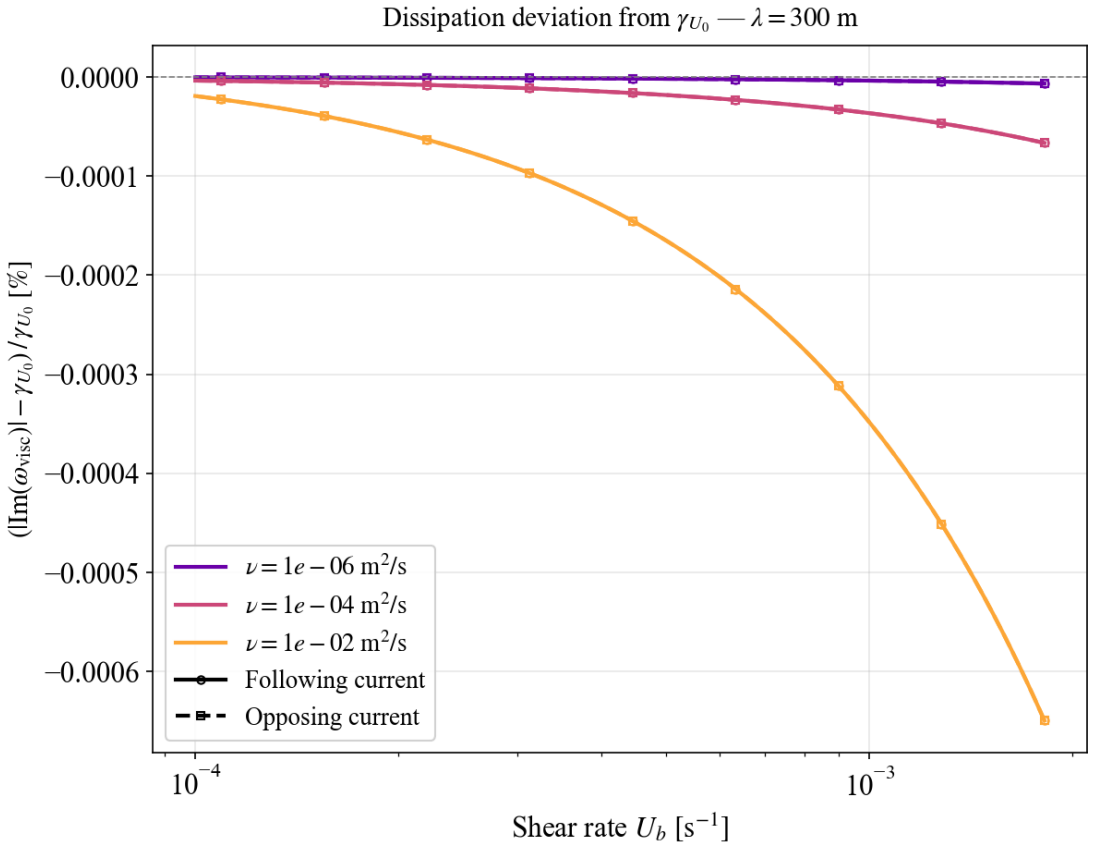


Figure 4.5.3: Relative dissipation-rate deviation for $\lambda = 300\,\mathrm{m}$, $(|\,\mathrm{Im}(\omega_{\mathrm{visc}})| - \gamma_{U_0})/\gamma_{U_0}$.

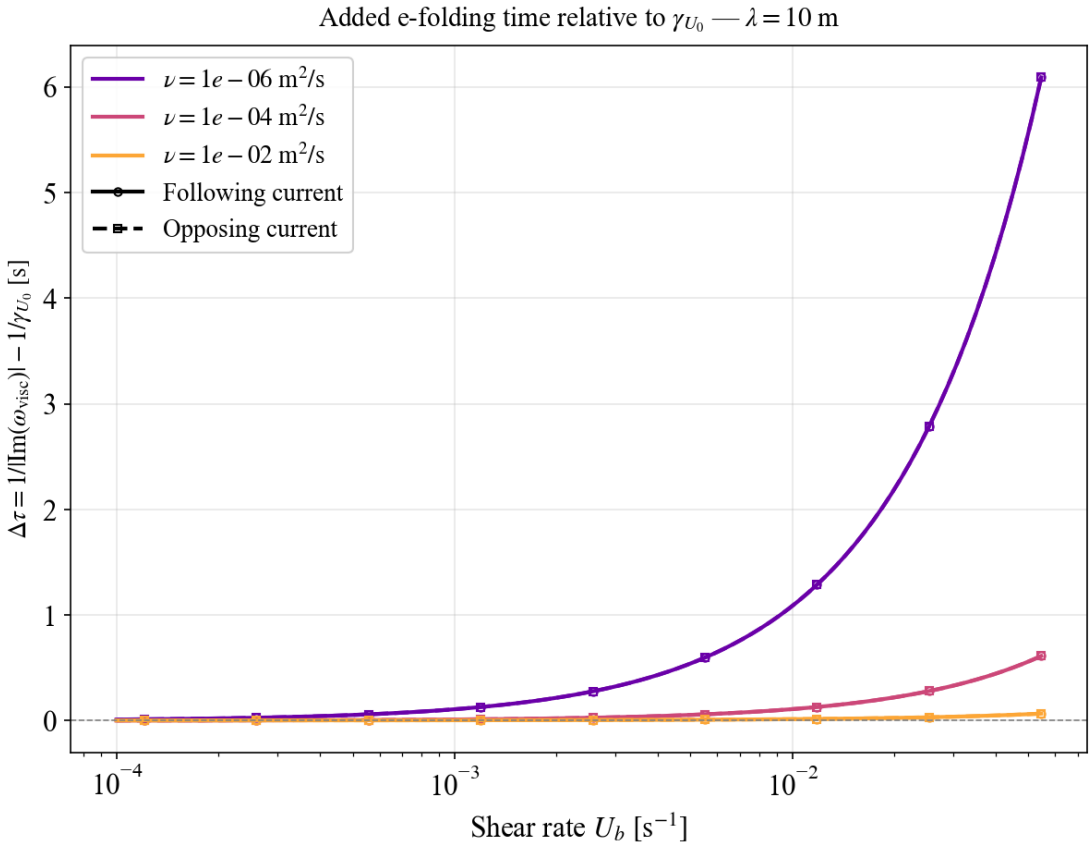


Figure 4.5.4: Additional e-folding time for $\lambda = 10\,\mathrm{m}$ induced by shear, $|\tau_{\mathrm{shear}}| - \tau_{U_0}$, where $\tau_{U_0} = \gamma_{U_0}^{-1}$, $\tau_{\mathrm{shear}} = \mathrm{Im}(\omega_{\mathrm{visc}})^{-1}$.

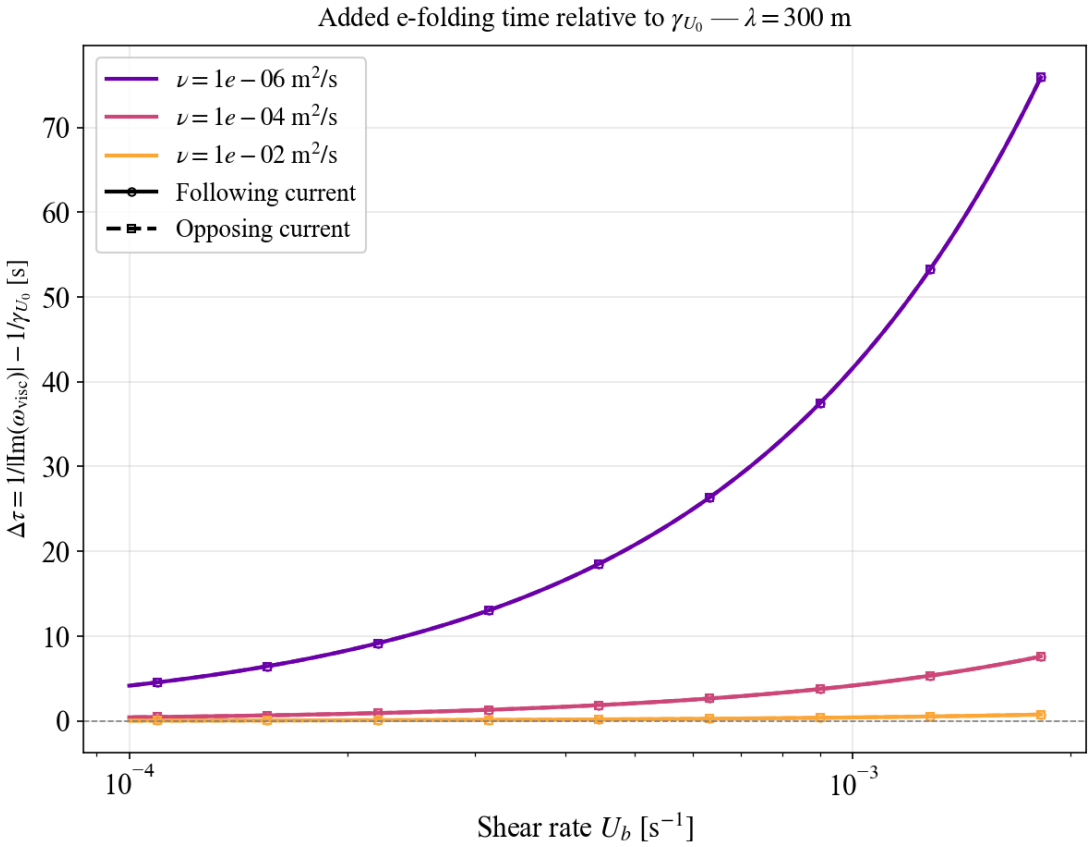


Figure 4.5.5: Additional e-folding time for $\lambda = 300\,\mathrm{m}$ induced by shear, $|\tau_{\mathrm{shear}}| - \tau_{U_0}$, where $\tau_{U_0} = \gamma_{U_0}^{-1}$, $\tau_{\mathrm{shear}} = \mathrm{Im}(\omega_{\mathrm{visc}})^{-1}$.

4.5.5 Reconstruction of the Physical Wave Amplitudes

As in the depth-invariant case, the velocity amplitudes follow directly from the solution for $w_0(z)$ obtained above, with $u_0(z) = (i/k)\,\partial_z w_0(z)$. The pressure amplitude $p_0(z)$, however, requires vertical integration of the momentum equation (4.1), which for the linearly sheared current does not yield a closed form and must be evaluated numerically. Unlike the depth-invariant case of §4.4, the reconstructed fields therefore offer no analytical expression that could serve as a ready-to-use input to the second-leading order. We consequently do not reconstruct the amplitudes here, noting that the loss of a closed form is a direct consequence of the depth-dependence introduced by the shear.

4.6 Second-Leading Order of WKB Expansion

Having solved the leading-order equations, we now expand the governing equations and boundary conditions (3.14) using the WKB setup (3.19). Reorganizing the equations and collecting the terms of order $\mathcal{O}(\mu)$, we find the following set of equations:

System of PDEs for WCI – Second-leading order

- **The horizontal momentum equation**

$$i\frac{k}{\rho}p_1 + w_1\frac{\partial U}{\partial z} - \left[i(\omega - kU) + \nu\left(\frac{\partial^2}{\partial z^2} - k^2\right)\right]u_1 = -\left[\frac{\partial u_0}{\partial \bar{t}} + U\frac{\partial u_0}{\partial \bar{x}} + u_0\frac{\partial U}{\partial \bar{x}} + W\frac{\partial u_0}{\partial z} + \frac{1}{\rho}\frac{\partial p_0}{\partial \bar{x}} - i\nu\left(2k\frac{\partial u_0}{\partial \bar{x}} + \frac{\partial k}{\partial \bar{x}}u_0\right)\right] \tag{4.90a}$$

- **The vertical momentum equation**

$$\frac{1}{\rho}\frac{\partial p_1}{\partial z} - \left[i(\omega - kU) + \nu\left(\frac{\partial^2}{\partial z^2} - k^2\right)\right]w_1 = -\left[\frac{\partial w_0}{\partial \bar{t}} + U\frac{\partial w_0}{\partial \bar{x}} + W\frac{\partial w_0}{\partial z} - w_0\frac{\partial U}{\partial \bar{x}} - i\nu\left(2k\frac{\partial w_0}{\partial \bar{x}} + \frac{\partial k}{\partial \bar{x}}w_0\right)\right] \tag{4.90b}$$

- **The incompressible continuity equation**

$$iku_1 + \frac{\partial w_1}{\partial z} = -\frac{\partial u_0}{\partial \bar{x}} \tag{4.90c}$$

- **The kinematic surface condition (at $z = 0$)**

$$w_1 = -i(\omega - kU)\eta_1 + \frac{\partial \eta_0}{\partial \bar{t}} + U\frac{\partial \eta_0}{\partial \bar{x}} - W \tag{4.90d}$$

- **Continuity of normal stress(at $z = 0$)**

$$p_1 = \rho g\eta_1 + 2\nu\rho\frac{\partial w_1}{\partial z} - 2\nu\rho W \tag{4.90e}$$

- **Continuity of tangential stress (at $z = 0$)**

$$ikw_1 + \frac{\partial u_1}{\partial z} = -\frac{\partial w_0}{\partial \bar{x}} \tag{4.90f}$$

- **The deep water condition (as $z \to -\infty$)**

$$u_1 \to 0, \quad w_1 \to 0 \tag{4.90g}$$

We will not be solving this system in this thesis, but we present the equations as they are the starting point of further research using the model built in §3.

Key Result 4.6: The second-leading order

We have derived the second-leading order system of equations of the WKB expansion in §3. This is an important and time saving contribution to further research of the WCI model derived in this thesis.

CHAPTER

FIVE

CONCLUSIONS & FUTURE WORK

5.1 Conclusions

This thesis has sought to develop a WCI framework that couples viscosity with linear surface waves and currents. The work has been conducted in the context of ocean conditions, for which WCI are always turbulent. To account for the turbulent effects, we have investigated the influence of viscosity on the model across a range spanning the molecular viscosity of water to a uniform eddy-viscosity parameterization. The uniform eddy-viscosity approximation is a very limited model, but gives some indications on how real turbulent effects affect WCI (Pope, 2000).

We solved the leading order of the derived WKB expansion for a depth-invariant background current and a linear shear current. The model reduced to known theory in the inviscid, no-current and no shear-limits, and produced corrections to the standard inviscid and dissipative theory for uniform eddy-viscosity parameterization. The corrections were much larger for the dissipative terms, than for the wave frequency. Corrections were also produced for molecular viscosity, but they were so small that they are likely negligible for all practical applications.

Returning to the research questions posed in §1.3, we present the key finding below as our answers:

1. **Reduction to the Orr-Sommerfeld equation:** Reducing the WKB leading-order governing equations to a single fourth-order ODE for the vertical velocity recovers the Orr-Sommerfeld equation.

2. **WCI dispersion relation for a depth-invariant current:** A closed-form dispersion relation was derived that recovers inviscid linear wave theory and the viscous Lamb damping rate at leading order in ν.

3. **Significant $\mathcal{O}(\nu^{3/2})$ correction to the dissipation rate:** The classical Lamb rate $\gamma_2 = 2\nu k^2$ remains accurate at molecular viscosity but deviates from the full dispersion relation by up to 3% in the eddy-viscosity regime;

retaining the γ_3 correction reduces this deviation to below 0.002%. Hence, $\gamma_{2,3}$ is preferred when using uniform eddy-viscosity parameterization, which have shown to correspond with observations under specific WCI conditions.

4. **WCI dispersion relation for a viscous linear shear current:** A dispersion relation coupling viscosity and linear shear was derived and shown to converge to the depth-invariant solution in the weak-shear limit and to inviscid shear theory in the inviscid limit.

5. **Negligible deviation from established theory validates the viscous linear-shear dispersion relation:** For realistic wind-induced current parameters, the derived relation yields only small and well-behaved corrections to established theory. The real frequency deviates from inviscid shear theory by below $\sim 10^{-2}\%$, and the shear correction to γ_{U_0} reaches a maximum of $\sim 0.05\%$. Because these corrections are physically consistent and negligible, they validate the relation as a framework unifying standard inviscid shear theory with viscous dissipation theory, although they reveal no new effect of the viscous-shear coupling. We hypothesise that such an effect may enter at the second-leading order and that shear corrections may become important under more advanced eddy-viscosity closures.

6. **Derivation of the second-leading-order system:** The second-leading-order WKB system was derived, providing the necessary groundwork for future work on amplitude modulation in the presence of viscosity.

5.2 Future work

For future research, the immediate next step is to solve the second-leading order, building on the leading-order solutions found in this thesis. These are the necessary input to (4.90), whose solution gives the WCI amplitude modulation under viscosity. The task is more demanding than at leading order, as the slowly varying variables $\bar{x}$ and $\bar{t}$ must now be treated explicitly.

A second direction builds on the compatibility established in §4.2.2. The leading-order equations admit a depth-dependent eddy viscosity, which addresses the limitation noted by Zhang et al. (2022), that eddy-viscosity closures validate least convincingly in the upper water column. Since the model converges with known results under the simplest eddy-viscosity parameterization, and viscosity and shear couples through the Airy function which decays extremely rapidly beneath the surface, we hypothesize that an eddy viscosity peaking near the surface at a higher value than applied in this thesis and decaying with depth enhances the viscous coupling. Confirming this would constitute a concrete improvement to WCI model accuracy.

Finally, the model could be coupled to larger-scale ocean and wind models. The Airy-type eigenmodes (4.72) are well suited here, and can be combined with a wind-stress boundary condition through which wind forcing enters.

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APPENDICES

APPENDIX

A

CLOSED FORM DISPERSION RELATIONS

Dispersion relations for depth-invariant $U = U_0$

$$\begin{aligned} A = &-128k^{12}\nu^6 + 16g^2k^6\nu^2 \\ &+ \frac{128}{3}\sqrt{\frac{1}{3}}\sqrt{11k^{12}\nu^6 - 18gk^9\nu^4 - g^2k^6\nu^2 - g^3k^3}\,k^6\nu^3 \\ &+ 4g^3k^3 - \frac{8}{27}\left(12k^4\nu^2 + gk\right)^3 + \frac{4}{3}\left(32k^8\nu^4 + 8gk^5\nu^2 - g^2k^2\right)\left(12k^4\nu^2 + gk\right) \end{aligned} \tag{A.1}$$

$$B = 192k^8\nu^4 + 16g^2k^2 + 12A^{1/3}gk + \frac{9A^{2/3}}{A^{1/3}} \tag{A.2}$$

$$\begin{aligned} \sigma_1 = -2ik^2\nu - \frac{1}{6}\sqrt{\begin{aligned}&24gk + \frac{4\left(96k^8\nu^4 + 24gk^5\nu^2 - 3g^2k^2 - \left(12k^4\nu^2 + gk\right)^2\right)}{A^{1/3}} \\ &- \frac{432\left(10ik^6\nu^3 + igk^3\nu - i\left(12k^4\nu^2 + gk\right)k^2\nu\right)}{\sqrt{B}} - 9A^{1/3}\end{aligned}} \\ - \frac{1}{6}\sqrt{B} \end{aligned} \tag{A.3}$$

$$\begin{aligned} \sigma_2 = -2ik^2\nu + \frac{1}{6}\sqrt{\begin{aligned}&24gk + \frac{4\left(96k^8\nu^4 + 24gk^5\nu^2 - 3g^2k^2 - \left(12k^4\nu^2 + gk\right)^2\right)}{A^{1/3}} \\ &- \frac{432\left(10ik^6\nu^3 + igk^3\nu - i\left(12k^4\nu^2 + gk\right)k^2\nu\right)}{\sqrt{B}} - 9A^{1/3}\end{aligned}} \\ - \frac{1}{6}\sqrt{B} \end{aligned} \tag{A.4}$$

$$\sigma_3 = -2ik^2\nu - \frac{1}{6}\sqrt{24gk + \frac{4\left(96k^8\nu^4 + 24gk^5\nu^2 - 3g^2k^2 - \left(12k^4\nu^2 + gk\right)^2\right)}{A^{1/3}} + \frac{432\left(10ik^6\nu^3 + igk^3\nu - i\left(12k^4\nu^2 + gk\right)k^2\nu\right)}{\sqrt{B}} - 9A^{1/3}} + \frac{1}{6}\sqrt{B} \quad \text{(A.5)}$$

$$\sigma_4 = -2ik^2\nu + \frac{1}{6}\sqrt{24gk + \frac{4\left(96k^8\nu^4 + 24gk^5\nu^2 - 3g^2k^2 - \left(12k^4\nu^2 + gk\right)^2\right)}{A^{1/3}} + \frac{432\left(10ik^6\nu^3 + igk^3\nu - i\left(12k^4\nu^2 + gk\right)k^2\nu\right)}{\sqrt{B}} - 9A^{1/3}} + \frac{1}{6}\sqrt{B} \quad \text{(A.6)}$$

Dispersion relations for linear $U = U_0 + bz$

$$\begin{aligned} A = {} & -128k^{12}\nu^6 + 16g^2k^6\nu^2 \\ & + \frac{32}{3}\sqrt{\frac{1}{3}}\sqrt{176k^{12}\nu^6 - 192iU_zk^{10}\nu^5 - U_z^4gk - 8U_z^2g^2k^2 - 16g^3k^3}\cdot k^6\nu^3 \\ & - 8\left(13U_z^2k^8 + 36gk^9\right)\nu^4 - 4\left(-13iU_z^3k^6 - 20iU_zgk^7\right)\nu^3 \\ & + \left(15U_z^4k^4 + 120U_z^2gk^5 - 16g^2k^6\right)\nu^2 \\ & - \left(iU_z^5k^2 + 24iU_z^3gk^3 + 80iU_zg^2k^4\right)\nu + 4g^3k^3 \\ & - \frac{1}{27}\left(24k^4\nu^2 - 8iU_zk^2\nu - U_z^2 + 2gk\right)^3 + 2\left(16k^8\nu^4 + 8gk^5\nu^2 + g^2k^2\right)U_z^2 \\ & + \frac{2}{3}\left(32k^8\nu^4 + 8gk^5\nu^2 - g^2k^2 - (4k^4\nu^2 + gk)U_z^2 - 8i(3k^6\nu^3 + gk^3\nu)U_z\right) \\ & \cdot\left(24k^4\nu^2 - 8iU_zk^2\nu - U_z^2 + 2gk\right) \\ & + 16i\left(8k^{10}\nu^5 + 6gk^7\nu^3 + g^2k^4\nu\right)U_z \end{aligned} \quad \text{(A.7)}$$

$$\begin{aligned} B = {} & \frac{1}{A^{1/3}}\Big(192k^8\nu^4 - 96iU_zk^6\nu^3 - 64U_z^2k^4\nu^2 + U_z^4 + 8U_z^2gk + 16g^2k^2 \\ & - 16\left(-iU_z^3k^2 - 4iU_zgk^3\right)\nu - 3A^{1/3}\left(-8iU_zk^2\nu - U_z^2 - 4gk\right) + 9A^{2/3}\Big) \end{aligned} \quad \text{(A.8)}$$

$$\begin{aligned} C = {} & -128ik^6\nu^3 - 64igk^3\nu + \left(8ik^2\nu + 2U_z\right)^3 \\ & - 8\left(24k^4\nu^2 - 8iU_zk^2\nu - U_z^2 + 2gk\right)\left(-4ik^2\nu - U_z\right) \\ & - 16\left(4k^4\nu^2 + gk\right)U_z \end{aligned} \quad \text{(A.9)}$$

$$D = 384k^8\nu^4 + 96gk^5\nu^2 - 12g^2k^2 - 12\left(4k^4\nu^2 + gk\right)U_z^2 - \left(24k^4\nu^2 - 8iU_zk^2\nu - U_z^2 + 2gk\right)^2 - 96i\left(3k^6\nu^3 + gk^3\nu\right)U_z \quad \text{(A.10)}$$

$$E = 1152k^4\nu^2 - 384iU_zk^2\nu + 18\left(8ik^2\nu + 2U_z\right)^2 - 48U_z^2 + 96gk \quad \text{(A.11)}$$

$$\sigma_1 = -2ik^2\nu - \frac{1}{2}U_z - \frac{1}{12}\sqrt{E + \frac{4D}{A^{1/3}} + \frac{27C}{\sqrt{B}} - 36A^{1/3}} - \frac{1}{6}\sqrt{B} \quad \text{(A.12)}$$

$$\sigma_2 = -2ik^2\nu - \frac{1}{2}U_z + \frac{1}{12}\sqrt{E + \frac{4D}{A^{1/3}} + \frac{27C}{\sqrt{B}} - 36A^{1/3}} - \frac{1}{6}\sqrt{B} \quad \text{(A.13)}$$

$$\sigma_3 = -2ik^2\nu - \frac{1}{2}U_z - \frac{1}{12}\sqrt{E + \frac{4D}{A^{1/3}} - \frac{27C}{\sqrt{B}} - 36A^{1/3}} + \frac{1}{6}\sqrt{B} \quad \text{(A.14)}$$

$$\sigma_4 = -2ik^2\nu - \frac{1}{2}U_z + \frac{1}{12}\sqrt{E + \frac{4D}{A^{1/3}} - \frac{27C}{\sqrt{B}} - 36A^{1/3}} + \frac{1}{6}\sqrt{B} \quad \text{(A.15)}$$

APPENDIX

B

NUMERICAL SOLVER FOR DEPTH-INVARIANT CURRENT

```
"""
Numerical solver for the viscous surface-gravity-wave dispersion relation.

Analytic damping-rate terms (intrinsic frequency expansion):
    gamma_2   =  2 nu k^2                                  [O(nu)]
    gamma_3   = -sqrt(2) nu^(3/2) k^3 / (g k)^(1/4)  [O(nu^(3/2))]
    gamma_2,3 =  gamma_2 + gamma_3

The full (squared) dispersion relation solved for the complex intrinsic
frequency sigma_a is

    -sigma_a^2 - 4 i nu k^2 sigma_a + 4 nu^2 k^4 + g k
        = 4 nu^2 k^3 sqrt(k^2 - i sigma_a / nu),

with damping rate gamma_exact = -Im(sigma_a).
"""

import numpy as np
from scipy.optimize import fsolve

g = 9.81


# ------------------------------------
# Analytic damping-rate terms
# ------------------------------------
def gamma_2(k, nu):
    """O(nu) Lamb damping rate."""
    return 2 * nu * k**2
```

```
30
31
32  def gamma_3(k, nu):
33      """O(nu^(3/2)) correction to the damping rate."""
34      return -np.sqrt(2) * nu**1.5 * k**3 / (g * k)**0.25
35
36
37  def gamma_23(k, nu):
38      """Corrected damping rate gamma_2 + gamma_3."""
39      return gamma_2(k, nu) + gamma_3(k, nu)
40
41
42  # ----------------------------------------------------
43  # Full dispersion relation: residual and root solver
44  # ----------------------------------------------------
45  def _disp_residual(xy, k, nu):
46      """Real/imag residual of the full dispersion relation
           for sigma_a = xy[0]+i xy[1].
47      Principal branch of the square root is enforced
           (Re(sqrt) >= 0)."""
48      sa  = xy[0] + 1j * xy[1]
49      sq  = np.sqrt(k**2 - 1j * sa / nu)
50      lhs = -sa**2 - 4j * nu * k**2 * sa + 4 * nu**2 * k**4 +
           g * k
51      rhs = 4 * nu**2 * k**3 * sq
52      r   = lhs - rhs
53      return [r.real, r.imag]
54
55
56  def solve_gamma_exact(k, nu_arr, rtol=1e-8,
       branch_rtol=0.05):
57      """Return damping rates gamma = -Im(sigma_a) by
           numerically solving the full
58      dispersion relation over an array of viscosities.
59
60      Continuation in nu (each solution seeds the next) keeps
           the root on the
61      surface-gravity-wave branch. Each root is asserted to be
           (1) converged,
62      (2) low-residual, (3) on the principal square-root
           branch, and (4) on the
63      surface-gravity-wave branch.
64      """
65      gamma    = np.empty(len(nu_arr))
66      sigma_r0 = float(np.sqrt(g * k))                  #
           inviscid real frequency
67      x0       = [sigma_r0, -float(2 * nu_arr[0] * k**2)]
68
69      for i, nu in enumerate(nu_arr):
70          sol, infodict, ier, msg = fsolve(
```

```
            _disp_residual, x0, args=(k, nu),
                full_output=True
        )

        # 1. fsolve reported convergence
        assert ier == 1, f"fsolve did not converge at
            nu={nu:.3e}: {msg}"

        # 2. residual small relative to characteristic LHS
            magnitude
        res   = np.hypot(*_disp_residual(sol, k, nu))
        scale = g * k
        assert res < rtol * scale, (
            f"large residual {res:.3e} (scale {scale:.3e})
                at nu={nu:.3e}"
        )

        # 3. principal branch holds for the converged root
        sa = sol[0] + 1j * sol[1]
        sq = np.sqrt(k**2 - 1j * sa / nu)
        assert sq.real >= 0, f"off principal branch at
            nu={nu:.3e}: Re(sqrt)={sq.real:.3e}"

        # 4. on the surface-gravity-wave branch
        assert sol[1] <= 0, f"non-physical growth (gamma<0)
            at nu={nu:.3e}"
        assert abs(sol[0] - sigma_r0) < branch_rtol *
            sigma_r0, (
            f"real frequency wandered off branch at
                nu={nu:.3e}: "
            f"sigma_r={sol[0]:.4f} vs
                sqrt(gk)={sigma_r0:.4f}"
        )

        gamma[i] = -sol[1]
        x0 = list(sol)

    return gamma


def solve_sigma_exact(k, nu):
    """Return the complex intrinsic frequency sigma_a for a
        single (k, nu).
    Real part is sigma_r; damping rate is -Im(sigma_a)."""
    x0  = [float(np.sqrt(g * k)), -float(2 * nu * k**2)]
    sol = fsolve(_disp_residual, x0, args=(k, nu))
    return sol[0] + 1j * sol[1]
```

Listing B.1: Solving the full dispersion relation for depth-invariant current.

APPENDIX

C

RECONSTRUCTION OF THE PHYSICAL FIELDS FOR DEPTH-INVARIANT CURRENT

For the leading order, we started out with a system of PDEs given by the variables (u_0, w_0, p_0, η_0), which were consequently reduced to a system of ODEs in terms of w_0. Having found the two eigenvalues λ_1, λ_2 and the relation between the coefficients A_1, A_2(which are related to the amplitude, but remains unspecified for the leading-order), we now have an equation for the vertical velocity amplitude $w_0(z)$, dependent on A_2 and the intrinsic frequency σ_a:

$$w_0(z) = \frac{i\sigma_a/\nu - 2k^2}{2k^2} A_2 e^{kz} + A_2 \exp\Big\{\sqrt{k^2 - i\sigma_a/\nu}\, z\Big\}. \tag{C.1}$$

Using (C.1) and the original set of equations, (4.1), we can thus find the solution to all the physical variables of the leading order. We find the horizontal velocity component via (4.1c):

$$u_0(z) = \frac{i}{k}\frac{\partial w_0(z)}{\partial z} = -\frac{\sigma_a/\nu + 2ik^2}{2k^2} A_2 e^{kz} + \sqrt{i\sigma_a/\nu - k^2} A_2 \exp\Big\{\sqrt{k^2 - i\sigma_a/\nu}\, z\Big\} \tag{C.2}$$

To reconstruct the pressure, we find the surface value combining (4.1d) and (4.1e) in terms of w_0 (effectively reducing the η-terms, and integrate (4.1a) over the system depth,

$$p_0(z) = p_0(0) + \int_0^z \frac{\partial p_0}{\partial z} dz = p_0(0) + \rho \int_0^z \left((i\sigma_a - k^2) w_0 + \nu \frac{\partial^2 w_0}{\partial z^2} \right) dz. \tag{C.3}$$

The integral gives

$$\int_0^z \frac{\partial p_0}{\partial z'} dz' = \rho\left(i\sigma_a - \nu k^2\right)\left(W_0(z) - W_0(0)\right) - ik\rho\nu u_0(z) + ik\rho\nu u_0(0) + C(\bar{x},\bar{t}) \tag{C.4}$$

where W_0 is the antiderivative of w_0,

$$W_0(z) = \left(\frac{i\sigma_a/\nu - 2k^2}{2k^3} e^{kz} + \frac{\exp\left\{\sqrt{k^2 - i\sigma_a/\nu}z\right\}}{\sqrt{k^2 - i\sigma_a/\nu}}\right). \tag{C.5}$$

The surface pressure is

$$p_0(0) = i\rho g\frac{w_0(0)}{\sigma_a} + 2\nu\rho\frac{\partial w_0}{\partial z}\bigg|_{z=0} = i\frac{\rho g}{\sigma_a} w_0(0) - 2ik\nu\rho u_0(0), \tag{C.6}$$

so that the full pressure equation is

$$p_0(z) = i\frac{\rho g}{\sigma_a} w_0(0) - ik\rho\nu(u_0(z) + u_0(0)) + \rho\left(i\sigma_a - \nu k^2\right)\left(W_0(z) - W_0(0)\right), \tag{C.7}$$

where W_0 is given as above and we have absorbed the integration constant into the free amplitude coefficient $A_2(\bar{x},\bar{t})$.

APPENDIX

D

AIRY FUNCTIONS

The Airy function is a special function satisfying the Airy equation,

$$\frac{d^2y}{du^2} - uy = 0, \tag{D.1}$$

which are both named after the British astronomer George Biddell Airy (b. 1801). The two linearly independent solutions to (D.1), Ai(u) and Bi(u), are oscillatory for $u < 0$, while for $u > 0$ Ai(u) is exponentially decaying and Bi(u) exponentially growing. Because of this change in character, we call $u = 0$ a turning point. Ai(u) and Bi(u), will henceforth be referred to as the Airy functions of the first and second kind, respectively.

In this thesis we will only use $Ai(u)$, as the deep water approximation requires decaying functions. For real values of u, Ai(u) is defined by the improper Riemann integral (Lakshminarayanan & Varadharajan, 2015):

$$\mathrm{Ai}(u) = \frac{1}{\pi}\int_0^\infty \cos\left(\frac{\tau^3}{3} + u\tau\right)d\tau. \tag{D.2}$$

The values of the Airy function of the first kind and its derivatives at the turning point $u = 0$ are known exactly, and given by

$$\mathrm{Ai}(0) = \frac{1}{3^{2/3}\,\Gamma\left(\frac{2}{3}\right)}, \quad \mathrm{Ai}'(0) = -\frac{1}{3^{1/3}\,\Gamma\left(\frac{1}{3}\right)}.$$

The Airy function can be extended to the complex plane, where, by Stokes' phenomenon, it admits asymptotic approximations as $|u| \to \infty$ that depend on $\arg(u)$. Later in this thesis, we will refer to these approximations as solutions valid along specific contours in the complex plane. For $|\arg(u)| < \pi$, Ai(u) is approximated by the asymptotic formula (Zwiebach, 2018):

$$\mathrm{Ai}(u) \sim \frac{1}{2\sqrt{\pi}} |u|^{-1/4} e^{-\xi}, \tag{D.4}$$

where

$$\xi \equiv \frac{2}{3}|u|^{3/2}.$$

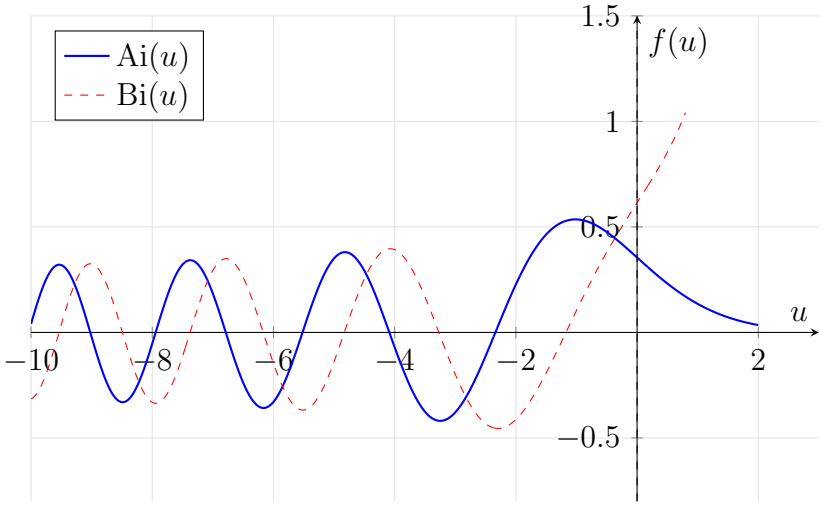


Figure D.0.1: The real-valued Airy functions of the first and second kind near the turning point $u = 0$.

APPENDIX

E

NUMERICAL SOLVER FOR LINEAR SHEAR CURRENT

```
1
2  """
3  Numerical solver for the viscous linear-shear dispersion
       relation (Eq. 4.86).
4
5  Profile: U(z) = U_0 + U_b z,  z in (-inf, 0].
6
7  The dispersion function D(omega)/Ai(u0) is evaluated along
       the local
8  steepest-descent contour in the complex u-plane. The
       physical integral
9      int_{-L}^{0} e^{k t} Ai(y(t)) dt
10 maps to (1/gamma) int e^{s u} Ai(u) du along a ray ending at
       u0; the entire
11 integrand lets the contour be deformed to the
       steepest-descent direction of
12 the exponent F(u) = s u - (2/3) u^{3/2}, giving monotone,
       non-oscillatory decay.
13
14 Float64 cancellation: near the root the integral and the
       boundary term cancel
15 by ~1e7x at small U_b and nu, limiting root accuracy to
       ~3e-12 rad/s. Where
16 Im(omega) is itself tiny, roots are polished with a 30-digit
       mpmath
17 complex-secant step (polish_root_mp). The same precision
       issue afflicts the
18 real-frequency correction Re(omega_visc) - omega_invisc, a
       difference of two
19 O(1) frequencies agreeing to ~16 digits; with_corr forms it
       directly in mpmath.
20 """
```

```
21
22 import numpy as np
23 from scipy.optimize import root
24 from scipy.special import airye
25 from scipy.integrate import quad
26
27 g = 9.81
28
29
30 # ----------------------------------------
31 # Inviscid baseline
32 # ----------------------------------------
33 def omega_invisc(k, U_0, U_b, g=9.81):
34     """Inviscid linear-shear dispersion relation
       (fixed-frame frequency)."""
35     return k * U_0 + np.sqrt(g * k + U_b**2 / 4) - U_b / 2
36
37
38 # ---------------------
39 # Parameter container
40 # ---------------------
41 class _AP:
42     def __init__(self, k, U_0, U_b, nu, g, L):
43         self.k = k; self.U_0 = U_0; self.U_b = U_b
44         self.nu = nu; self.g = g; self.R = U_b / nu; self.L
           = L
45
46
47 # ------------------------------------------------------------
48 # Dispersion function (float64, steepest-descent contour)
49 # ------------------------------------------------------------
50 def _surf(omega, p):
51     """D(omega)/Ai(u0) along the steepest-descent contour.
52
53     The /Ai(u0) normalization cancels the common exp(phi0)
       scale of all three
54     terms, keeping the function O(1) for root-finding at
       small U_b.
55     """
56     sig0  = omega - p.k * p.U_0
57     alfa  = 1j * (p.g * p.k**2 / sig0 - p.k * p.U_b)
58     beta  = 3 * p.nu * p.k**2 - 1j * sig0
59     c     = (omega / p.k - p.U_0) / p.U_b
60     coeff = (np.exp(1j * 5 * np.pi / 6) if p.U_b > 0 else
       np.exp(1j * np.pi / 6)) \
61           * np.cbrt(p.k * p.R)
62     s     = p.k / coeff
63     u0    = coeff * (-c - 1j * p.k / p.R)      # = y(0)
64
65     ai0, aip0, _, _ = airye(u0)
66     phi0 = (2.0 / 3.0) * u0**1.5
```

```
67
68      # Steepest-descent direction away from u0 (F'(u0) = s -
            sqrt(u0)).
69      Fp = s - np.sqrt(u0)
70      e  = (-Fp.conjugate()) / abs(Fp)
71      # Truncate after ~40 e-foldings, but never beyond the
            image of the
72      # physical contour length |gamma|*L.
73      Rmax = min(abs(coeff) * p.L, 40.0 / abs(Fp))
74
75      def integrand_u(r):
76          u = u0 + e * r
77          aie, _, _, _ = airye(u)                 #
                Ai(u)*exp((2/3)u^{3/2})
78          expo = s * (u - u0) - (2.0 / 3.0) * u**1.5 + phi0
79          if expo.real > 500:
80              return 0.0j
81          return np.exp(expo) * aie
82
83      jr, _ = quad(lambda r: integrand_u(r).real, 0.0, Rmax,
84                   limit=200, epsabs=1e-12, epsrel=1e-12)
85      ji, _ = quad(lambda r: integrand_u(r).imag, 0.0, Rmax,
86                   limit=200, epsabs=1e-12, epsrel=1e-12)
87      intv = -(e / coeff) * (jr + 1j * ji)
88
89      return (2 * p.k**2 * (p.nu * p.k**2 - beta) * intv
90              + (alfa + beta * p.k - p.nu * p.k**3) * ai0
91              + 2 * p.k**2 * p.nu * aip0 * coeff) / ai0
92
93
94  def _airy_res(x, p):
95      om = x[0] + 1j * x[1]
96      r  = _surf(om, p)
97      return [r.real, r.imag]
98
99
100 #
        ----------------------------------------------------------------
101 # mpmath polish (high-precision refinement where float64
        cancels)
102 #
        ----------------------------------------------------------------
103 POLISH_IM_THRESHOLD = 1e-7      # polish roots with
        |Im(omega)| below this [rad/s]
104 POLISH_RE_CORR_ABS  = 1e-7      # polish when |Re(omega_visc)
        - omega_invisc| below this
105
106
107 def polish_root_mp(omega64, k, U_0, U_b, nu, g=9.81, L=3.0,
        dps=30,
108                    with_corr=False):
```

```
    """30-digit complex-secant refinement of a float64 root.

    The dispersion function is re-evaluated in mpmath (plain Ai-ratios are
    stable at high precision; no exponential scaling needed). Converges in
    2-3 steps. Falls back to the input root if mpmath is unavailable or the
    iteration stalls. If with_corr, also returns the real-frequency viscous
    correction Re(omega_visc) - omega_invisc formed entirely in mpmath,
    avoiding the catastrophic float64 cancellation of subtracting two O(1)
    frequencies that agree to ~16 digits.
    """
    try:
        from mpmath import (mp, mpf, airyai, exp as mexp, sqrt as msqrt,
                            quad as mquad, cbrt, pi as mpi, conj, fabs)
    except ImportError:
        if with_corr:
            return omega64, omega64.real - omega_invisc(k, U_0, U_b, g)
        return omega64

    mp.dps = dps
    _m = lambda x: mpf(repr(float(x)))
    k_, U0_, Ub_, nu_ = _m(k), _m(U_0), _m(U_b), _m(nu)
    g_, L_ = _m(g), _m(L)
    R_ = Ub_ / nu_

    def D(om):
        sig0  = om - k_ * U0_
        alfa  = 1j * (g_ * k_**2 / sig0 - k_ * Ub_)
        beta  = 3 * nu_ * k_**2 - 1j * sig0
        c     = (om / k_ - U0_) / Ub_
        coeff = (mexp(1j * 5 * mpi / 6) if Ub_ > 0
                 else mexp(1j * mpi / 6)) * cbrt(k_ * R_)
        s     = k_ / coeff
        u0    = coeff * (-c - 1j * k_ / R_)
        ai0   = airyai(u0)
        Fp    = s - msqrt(u0)
        e     = (-conj(Fp)) / fabs(Fp)
        Rmax  = min(fabs(coeff) * L_, 50 / fabs(Fp))
        f     = lambda r: mexp(s * e * r) * airyai(u0 + e * r) / ai0
        intv  = -(e / coeff) * mquad(f, [0, Rmax])
        return (2 * k_**2 * (nu_ * k_**2 - beta) * intv
                + (alfa + beta * k_ - nu_ * k_**3)
```

```
                + 2 * k_**2 * nu_ * airyai(u0, 1) / ai0 *
                    coeff)

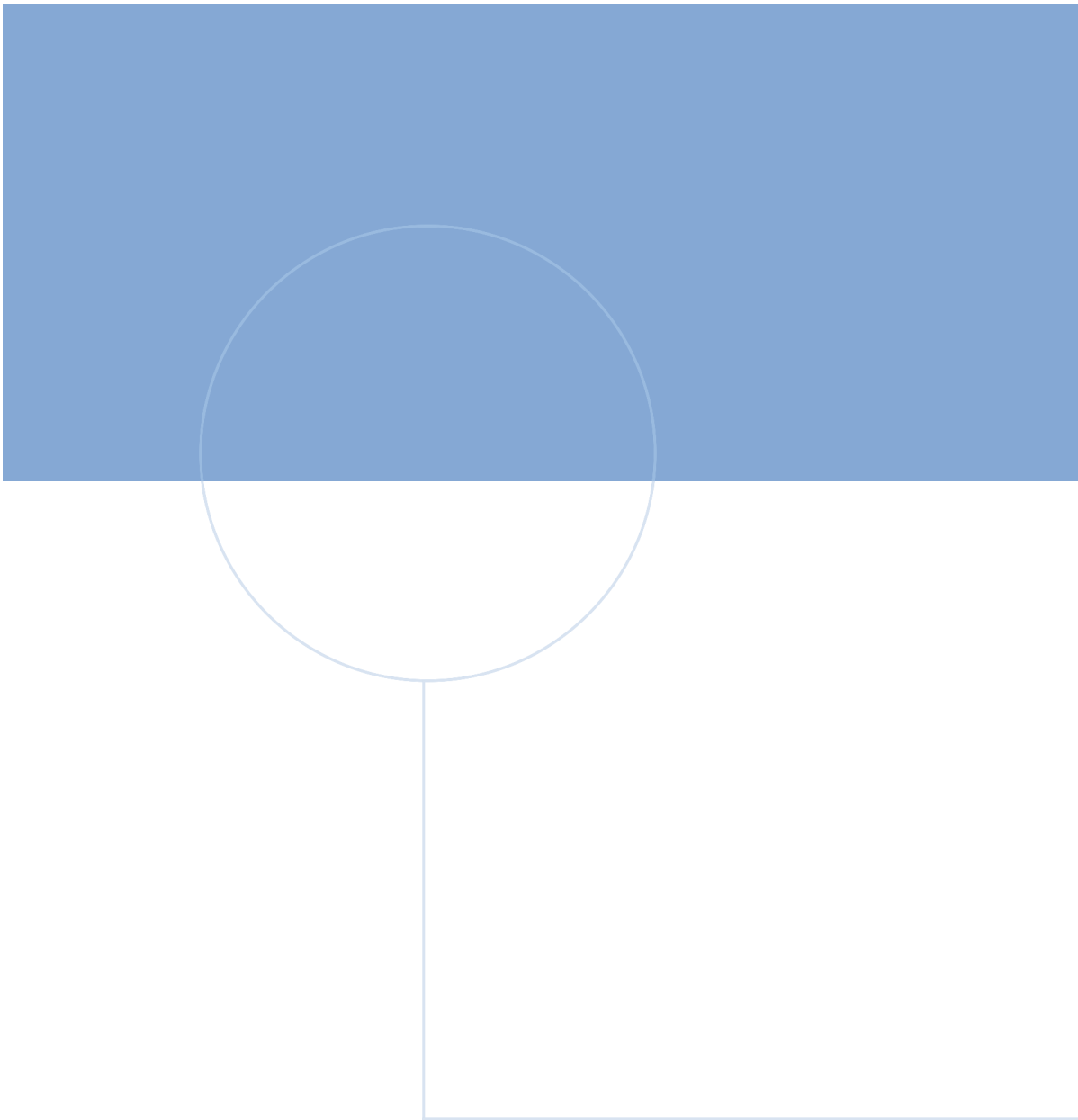
    om0 = mp.mpc(omega64)
    om1 = om0 * (1 + mpf("1e-9"))
    D0, D1 = D(om0), D(om1)
    for _ in range(12):
        if D1 == D0:
            break
        om2 = om1 - D1 * (om1 - om0) / (D1 - D0)
        om0, D0 = om1, D1
        om1, D1 = om2, D(om2)
        if fabs(D1) < mpf("1e-25"):
            break
    if with_corr:
        inv_mp = k_ * U0_ + msqrt(g_ * k_ + Ub_**2 / 4) -
            Ub_ / 2
        return complex(om1), float((om1 - inv_mp).real)
    return complex(om1)


# ----------------------
# Top-level root solve
# ----------------------
def ana_visc_shear(k, U_0, U_b, nu, g=9.81, L=3.0,
    omega0=None, polish=True):
    """Solve the viscous linear-shear dispersion relation
        for complex omega.

    omega0 seeds the root finder (use the previous root for
        continuation in
    U_b); defaults to the inviscid root plus the Lamb
        dissipation rate, which
    selects the correct branch even at strong shear. Returns
        the complex
    fixed-frame frequency, or None if both solve attempts
        fail.
    """
    p = _AP(k, U_0, U_b, nu, g, L)
    if omega0 is None:
        omega0 = omega_invisc(k, U_0, U_b, g) - 2j * nu *
            k**2
    sol = root(_airy_res, [omega0.real, omega0.imag],
        args=(p,),
                method='hybr', tol=1e-11)
    if not sol.success:
        sol = root(_airy_res, [omega_invisc(k, U_0, U_b, g),
            -2 * nu * k**2],
                   args=(p,), method='hybr', tol=1e-11)
        if not sol.success:
            return None
```

```
    om = complex(sol.x[0], sol.x[1])
    if polish and abs(om.imag) < POLISH_IM_THRESHOLD:
        om = polish_root_mp(om, k, U_0, U_b, nu, g, L)
    return om
```

Listing E.1: Solving the viscous linear shear dispersion relation



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